

SMART CANE FOR VISUALLY IMPAIRED

A project report submitted in partial fulfillment
of the requirements for the degree of
Bachelor of Technology
in
Electronics & Computer Engineering

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Abstract

The Smart Cane is an assistive device designed to improve the mobility and safety while navigating for visually impaired individuals. It integrates real time obstacle detection using ultrasonic sensors, fall detection using motion processing unit, GPS based turn navigation and live location tracking, and acts as a support system by notifying caretakers by sending them emergency alerts if the user falls. The cane detects nearby obstacles and provides haptic feedback, ensuring safe movement. The cane monitors the user's stability using gyroscope and accelerometer, detects falls and notifies the caretakers through a mobile application when the user has fallen. The caretakers can monitor the user's real-time live location through the mobile app, while turn-by-turn navigation helps the user in navigating to different destinations by providing audio instructions. The cane is connected to firebase cloud database, so that data can be exchanged between the cane and mobile application seamlessly. The integration of all these features ensures a user-friendly experience, which promotes independence and confidence to the user while navigating.

This project focuses on the practical implementation of assistive technology to improve accessibility and safety for visually impaired individuals. By making use of using IoT and cloud-based technologies, the Smart Cane provides real time feedback and remote monitoring, which reduces the gap between the users and caregivers. The cane boosts mobility for the user, lowers chances of falling accidents significantly, and saves the user from being dependent on someone else or a guide dog to help the person move without assistance. This unique approach gives evidence for how smart technologies can be made to include solutions for disabled individuals. The Smart Cane can greatly enhance independent mobility and safety for all visually impaired individuals.

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Chapter 1

Introduction

Mobility is a basic constituent of human life, encompassing independence, participation in society, and the overall well-being of persons. For individuals with visual impairment or physical hindrances related to old age, the journey of independent mobility is steeped with challenges that often hamper self-confidence and safety. Conventional aids, such as walking canes, do serve their purpose through tactile stimulation for the detection of some obstacles nearby, but they are limited by design. These cannot provide real-time feedback about the surroundings or the events happening therein beyond mere contact: there may be dynamic situations such as moving objects, changing terrains, or sudden health stipulations-alike the falling of a person. In an age where technology becomes the lifeline, there is huge scope of converting the traditional assistive devices into intelligent interactive systems giving real-time, physical, and digital support to the user. By bringing in embedded system technologies, wireless communication technologies, sensor technology, and cloud integration, very basic assistance of mobility can now be changed into an intelligent adaptive system to cater to these needs and inherently secure the users' protection.

One such characteristic of intelligent systems is the capability regarding sensing and understanding the user's surroundings and physical status. Through environmental sensing techniques, the system senses probable hazards like walls, poles, uneven surfaces, or moving objects before a user comes across them physically. It gives much higher safety and self-confidence. At the same time, it keeps a track of body movement which consists of an acceleration, orientation, and stability with respect to the user. This allows systems to detect certain events such as slips or falls and to make appropriate responses against them. Most elderly users would find this feature useful because it prevents a delay in medical attention due to the harmful nature of a fall. Combining environmental and motion sensing creates a system that understands both the world around the user and the user's condition, enabling smart responses in different situations.

Navigation is another critical challenge for individuals who cannot rely on vision to move through spaces. Finding the right path, avoiding obstacles, and maintaining direction are especially difficult in unfamiliar areas. Traditional solutions such as memorizing routes or relying on guidance from others can be unreliable. Modern navigation technologies provide a solution by using real-time location tracking and digital route planning to guide users to their destination. When navigation instructions are provided through audio feedback, they become fully accessible to visually impaired users. These instructions allow the user to follow a route step-by-step without the need to see a screen or ask for directions. Caregivers can also assist remotely by setting destinations or tracking the user's route using a mobile application, making the system more supportive and flexible.

Communication serves the primary purpose of bolstering user safety and confidence. It enables the exchange of salient data in real time when the smart device is connected to the Internet. For example, if a fall is detected by the system, a notification will be sent with the user's current location to immediately alert a caregiver. The caregiver can also track the user's movement through the app, including their location and progress along the route. Such communication ensures that a user is never really alone, even when they are physically alone. Emergency functions such as alerts or remote guidance can also be activated as required. In this way, the smart system functions not only as a mobility aid but also as a safety monitor and a communication line between the user and their caregiver.

The design and development of such systems involve the integration of multiple technologies. Embedded systems handle local processing, sensor data is interpreted using logic-based algorithms, and wireless communication modules send data to and from cloud platforms. The mobile application acts as a user interface for caregivers, enabling interaction with the system. All components must be carefully designed to operate together in real time with limited power and space. Since this system is used daily and in different conditions, it also needs to be comfortable, portable, and simple to use. The goal is not only to add functionality but to make the technology feel natural and helpful to the user.

This project focuses on the design and development of a Smart Cane system that brings together modern technologies to provide intelligent support for visually impaired and elderly individuals. It integrates features such as obstacle detection, fall recognition, step-by-step navigation using audio feedback, and a real-time connection to a mobile application for caregivers. The system collects and processes data about the environment and user movement, enabling timely responses to hazards or emergency situations. Caregivers can track the user's location, receive instant alerts during a fall, and send navigation routes remotely through the app. All of this is made possible through wireless communication and cloud-based data management. Designed to be user-friendly, portable, and power-efficient, the Smart Cane aims to overcome the limitations of traditional mobility aids by actively assisting, guiding, and protecting users. More than just a walking aid, it acts as a digital companion that enhances safety, builds independence, and connects users to those who care for them.

Chapter 2

Literature Survey

S. Kumar et al. [1] proposes a CNN-based Smart Cane for navigation and obstacle detection for visually impaired people. This real-time object detection and ultrasonic sensor supplement spatial awareness uses CNNs. The focus of research work is on integrating AI-driven models in embedded systems addressing challenges such as computation efficiency, real-time processing, and low-power consumption. It provides an auditory or haptic signal feedback to the user for navigating within a particular surrounding. The smart cane was tested in real conditions where it has shown a major improvement in obstacle detection and avoiding compared to traditional mobility aids. The study emphasizes the future potential of deep learning along with sensor fusion in assistive technology which will open future works in this field of smart mobility solutions for the visually impaired.

S. Khan et al. [2] provides a systematic review that analyzes different navigation assistants for the blind and visually impaired: electronic travel aids, smartphone-based navigation systems, and the most current and improving wearable devices. It has evaluated mobility effectiveness with respect to these solutions and identified key challenges, including obstacle detection, real-time navigation, and user adaptability. Based on their design, navigation assistants can be classified into vision-based, sensor-based, and hybrid systems, each having their pros and cons. Vision-based systems make use of cameras and deep learning models to interpret surroundings, while sensor-based systems use obstruction detection mechanisms such as LiDAR, ultrasonic, and infrared sensors. Hybrid models attempt to integrate the two technologies to be more reliable but cannot be realized due to high-power consumption, processing latency, or even the cost. User feedback on existing navigation systems is also discussed, focusing on some critical aspects such as easy understanding of the interface and real-time reaction. Possible developments like artificial intelligence, edge computing, and energy-efficient hardware will help in making these systems even more applicable to attract people in day-to-day life. While a lot of advancements have come to existence where noticeable improvements have been achieved in the mobility of the visually impaired, the review emphasizes the necessity for increased research and innovation for providing better adaptive navigation solutions.

K. Jivrajani et al. [3] describe the design of a smart stick intended for the use of blind people, along with an incorporation of AI and IoT. The paper mentions the combination is to enhance user mobility and safety in terms of real-time obstacle detection, navigation assistance, and alarming in emergencies. The proposed solutions rely on AI algorithms to provide object recognition; the extra information received allows the stick to give the user accurate aid compared with other systems and solutions. IoT makes it possible to monitor and communicate remotely so caregivers can follow a user's movements and react to emergencies. Experimental tests demonstrated the efficiency of the system, including timely warnings and detection of obstacles with almost 100% accuracy. The authors emphasize the feasibility of an AIoT to assist in the transformation of assistive technologies for navigation into a more accessible and reliable form for the visually impaired.

W. -J. Chang et al. [4] proposes that an artificial intelligence-based edge computing will provide an assistive system to make zebra crossings safer for a visually impaired pedestrian. The system proposed highly integrates deep learning and image-processing techniques—specifically CNNs—to detect pedestrian signals, crosswalk patterns, and even the approach of vehicles accurately. Unlike traditional models using cloud computing that suffer by high latency and permanent connections to the internet, this is an edge computing model that processes data locally for real-time decision-making feedback. The author has identified several important challenges like computational efficiency, environmental changes, and accuracy with at least two lighting and weather conditions. To cover this, it has been optimized for lightweight AI processing to be applied in real life, and voices would include field tests. The system had a very high rate of accuracy in the detection of safe crossing opportunities and reduced reaction time to crossing options for the visually impaired inputted from real-world datasets. Using auditory or haptic feedback, this innovation significantly improves mobility and independence for the visually impaired community. The conclusions provide amongst the best promises for paths in assistive navigation and pedestrian safety, rendering urban spaces more possible and accessible.

T. Lavanya Narayani et al. [5] put forth the idea of integrating a Smart Cane with a camera module aimed at enriching mobility for the visually impaired. The system employs image processing techniques for obstacle detection and object recognition for those things outside their immediate environment. It aims to use a camera-based approach to overcome the limitations of conventional obstacle-wise detection by sensor-based smart canes. Integrating a microcontroller with image processing algorithms for locating objects and alerting its users through auditory feedback has enhanced real-time navigation for the model. The performance assessment in several real field situations was able to show great advancement in the accuracy of obstacle detection and usability. It was surmised that combining sensor-based technology with vision-based technologies could lead to making a more suitable assistive device for the visually impaired, as mentioned in this research study.

A new methodology proposed by **T. P. Viet et al. [6]** integrates obstacle detection into assistive devices with the Random Forest algorithm. The system facilitates movement for the visually impaired by early detection of obstacles and giving real-time feedback. Such model will integrate machine learning techniques to classify static and dynamic obstacles using different sensor inputs for improvement toward better detection accuracy than traditional ultrasonic-based methods. Moreover, using a dataset of various scenarios encountered concerning obstacles, Random Forest classifier optimizes the making of decisions along reducing false positives. The experiment would be conducted under various environmental conditions to measure the performance of the present system while showing robustness in real-field environments. The experimental results demonstrate how efficient the model stands in detecting static and dynamic obstacles along with perfect navigation assistance. These findings imply that machine learning-assisted approaches would benefit the efficacy of the mobility aids, making them smarter and more responsive.

R. Ghatkamble et al. [7] developed a smart real-time navigation system and object identification for visually impaired persons via computer vision and IoT. The system has as a part wearable device carrying a camera for surroundings capture. The surrounding environment is captured and processed using a Convolutional Neural Network (CNN) for the purpose of object detection as well as an Optical Character Recognition (OCR) engine for reading text from labels and signs. This information is then uploaded to the cloud infrastructure with the IoT-based approach that allows communication from the wearable device to the output module. Real-time feedback channel is accessed through the text-to-speech module, which speaks detected objects and texts as information to the user. The system is augmented as an edge computing framework to provide response latency and trade-off computation efficiency. The findings exude a system competitive in window detection and recognition of optical characters and hence improved knowledge and autonomy in travel for visually impaired people.

A. Supekar et al. [8] introduces a Portable Navigation System. Based on this feature, the system is designed specifically for helping disabled persons traverse their immediate environments. Component features include image recognition, text-to-speech conversion and audio feedback designed to increase understanding. The model uses optical character recognition (OCR) to extract text from the images thus enabling the visually impaired persons to interpret various signboards and other written information in their surroundings. The extracted text is also converted to audio output through the speech synthesis technique for real-time assistance. One of the critical components in the model is camera-based object detection for obstacle and landmark identification to enhance safer navigation during the user's movements. The system is portable and cover small volume, allowing daily portability. The performance results demonstrate the potential of this portable device to provide accurate text recognition with clear voice output, thereby increasing the independence of visually impaired individuals. This integrated model, built up from various assistive technologies, shows the real solution against many navigation problems encountered by disabled users.

S. Zafar et al. [9] proposes a smart cane system integrating LiDAR and ROS for the sake of improving mobility for visually impaired individuals. The system uses LiDAR sensors for real-time environmental mapping to accurately detect obstacles and estimate distance. ROS provides the means for sensor fusion and data processing, thereby consolidating navigation. The cane is continuously scanning for obstacles within varying distances and angles, giving real-time feedback to the user. In contrast with conventional smart white cane systems, the proposed system offers an improved level of spatial awareness, adaptation to dynamic environments, and safety assurance. The results of simulations and real-time experiments have proved that this particular system is enhancing navigation, thus being an apposite assistive technology for blind individuals.

M. A. Khan et al. [10] proposes a smart cane that is portable and works with the Android platform. This smart cane is known to be the one with ultrasonic sensors attached to detect obstacles, and it integrates a smartphone app that provides real-time feedback to a wearer's movement. This cane communicates automatically with the Android device via Bluetooth, where the app will process the sensor data so that it can give auditory or vibration alerts. The system is highly portable, and at the same time, it is easy to use. In their design, it has proved to yield better performance as an obstacle avoider when compared to the traditional white cane. The mobile app integration also allows the users to set personalized feedback settings that make navigation safer and more intuitive. The test results demonstrate the system's effectiveness in both indoor and outdoor environments and show substantial improvements in spatial awareness as well as in user safety.

Q. Na et al. [11] presents a novel indoor navigation system for the visually impaired that enhances path generation and incorporates human bipedalism constraints for smooth and natural movement, ensuring. It gathers real-time information on the surroundings from sensors and uses mapping techniques to provide optimum paths while avoiding obstacles. The navigation algorithm also generates routes considering factors like human gait pattern, length of step, and stability of walking in accordance with natural walking dynamics. It is adaptable; hence, it changes the path with movements of the user, minimizing sharp changes in direction and thereby promoting spatial awareness. This system delivers guidance as compared to conventional aids, thus assisting with increased mobility and safety in indoor settings.

A smart voice navigation and object perception system to support visually impaired persons for real-time lectures on navigation and object recognition is proposed by **S. R. Kawale et al. [12]**. Voice guidance is coupled with sophisticated sensor technologies regarding obstacle avoidance and environmental object identification. The classification of objects is done using computer vision and deep learning techniques, which then convert the results to a location by means of spoken language. For broader detection of obstacles keeping a safe way to safely navigate, the system uses ultrasonic sensors along with LiDAR. This method allows for voice-based navigation to be interactive and intelligent compared to regular assistive tools and gives the visually impaired greater spatial awareness and proprioception towards independent movement.

N. Druml et al. [13] presents a virtual white cane that utilizes Time-of-Flight (ToF) 3D imaging technology which detects the surrounding area and helps blind and visually impaired persons in their mobility. Unlike white canes that use physical contact where they get in sight of obstacles at various distances from the tip and provide real-time acoustic or haptic feedback, this system uses 3D depth sensing to detect obstacles. This ToF sensor allows measuring the depth with high precision, increasing spatial awareness and obstacle avoidance. The system, therefore, makes navigation more instinctive and effective by removing most of the limitations of conventional mobility aids. Compared to existing smart cane solutions, this offer improvement concerning accuracy, detection time, and flexibility with changing environments.

B. Veerasamy et al. [14] proposes an intelligent assistive cane that optimizes navigation and obstacle detection for people with visual impairment through edge-cloud collaboration. Their architecture featured edge computing for online data processing and cloud services for both advanced analysis and facility for remote assistance. It enhances that detection, measurement of real time feedback, either auditory or haptic, by a variety of sensors: ultrasonic and LiDAR. Ultimately, object recognition and mapping of the environment can be done in the cloud, resulting in reduction in the latency for efficient data transmission through this edge-cloud architecture. When compared to traditional and stand-alone smart canes, this intelligent solution increases accuracy and responsiveness at the same time as it extends the general scope of the implementation options the intelligent cane has for the visually-impaired.

V.K.K. et al. [15] presented an extensive account of the advances in the already-classified innovations and the current trends associated with smart cane technologies for the visually challenged. Smart canes are equipped with ultrasonic, LiDAR, infrared, and Time-of-Flight sensors for significant enhancement of user detection and obstacle awareness, given their complement of navigation assistance and real-time feedback. Emerging technology such as artificial intelligence (AI) and Internet of Things (IoT) in combination with edge-cloud augmentation introduced real-time processing of data, object recognition, and environmental mapping. Besides, many studies concentrate on the multimodal feedback system that adds auditory, haptic, and voice-based alerts to complement the interaction. Compared with the standard white cane, the smart cane has better safety, autonomy, and adaptation in dynamic environments. Future directions in wearable integration, AI-driven navigation, and cloud-based assistive services to improve accessibility and mobility while addressing challenges such as power consumption, affordability, and the limitations in real-time processing is highlighted in this review.

The Digital Toolkit developed by **A. Koti et al. [16]** is a kind of approach suggested to improve the mobility of sightless people. It has provision for obstruction and object detection with distance measurement, currency identification, traffic sign recognition, and battery status updates every minute. It can be easily worn, mounted, or carried around without assistance; has the capacity for solar recharging; and identifies known faces, strangers (with alarms), and animals. It chiefly comes in the form of a "DIGI white Cane" which could provide greater autonomy and navigation for the blind and those having low vision.

Chapter 3

Methodology

3.1 System Overview

The Smart Cane is a mobility assistive device that aims at increasing the independence and safety of people with visual impairments. The device is equipped with functions such as real-time obstacle detection, fall detection function, turn-by-turn navigation, and emergency alerts regarding all types of outdoor support. The system partners with a mobile application Cane Connect for all its functionalities using a Firebase Cloud Database to ensure it fully connects with an IoT-enabled device.

The heart of the system is the ESP32 microcontroller, whose mission is to process the data collected from sensors and communicate with cloud services. As per orientation, the cane is attached with three ultrasonic sensors captured on both sides to detect objects in various directions; these sensors offer haptic feedback to the user to indicate the presence of the object. Further, for fall detection, falls are detected by an MPU6050 sensor which comprises a 3-axis accelerometer and a 3-axis gyroscope. The motion detector is basically measuring the orientation throughout the falls, and if it detects a fall as per certain conditions, it is updated with the Firebase immediately and sent as a notification to the caregiver through the mobile application.

A GPS module continuously enables a person in navigation to have updated information on their location. The generation of direction instructions is done by the Google Maps Directions API, while that of audio WAV files is done by Android Text-to-Speech (TTS). These files are subsequently uploaded to Firebase Storage and played back through a MAX98357A I2S DAC module connected to a speaker, giving clear sound guidance. The same system monitors and updates the starting and end points of navigation sessions on Firebase to keep the caregiver informed about the user's travel status.

Mobile Application & Cloud Integration:

The Smart Cane is supported by a companion mobile application that provides real-time monitoring and remote assistance. The app connects to Firebase, enabling caregivers to:

- Track the user's live location on a map using GPS data from the cane.
- Receive fall alerts when a fall is detected.
- Send navigation instructions to the cane remotely, which are then converted to audio using the Android Text-to-Speech (TTS) API and sent as links to the audio files stored in Firebase.
- The app keeps track of the Smart Cane power status. Displays whether the cane is online or offline in the mobile app.

The system is powered by rechargeable Li-ion batteries, with an TP4056 charging module for efficient power management. The integration of sensor-based feedback, cloud connectivity, and a user-friendly mobile app makes the Smart Cane a complete mobility aid for visually impaired individuals, significantly enhancing their independence and safety.

3.2 System Architecture

The Smart Cane system is designed with a well-structured architecture that integrates multiple components to assist visually impaired individuals. It consists of three primary modules:

- i) **Smart Cane Hardware:** The physical device equipped with sensors, a microcontroller, and a speaker for audio feedback.
- ii) **Firestore Cloud Backend:** A cloud-based real-time database that facilitates communication between the Smart Cane and the mobile application.
- iii) **Cane Connect Mobile Application:** An interface for caregivers to monitor the user's status, send navigation instructions, and receive alerts.

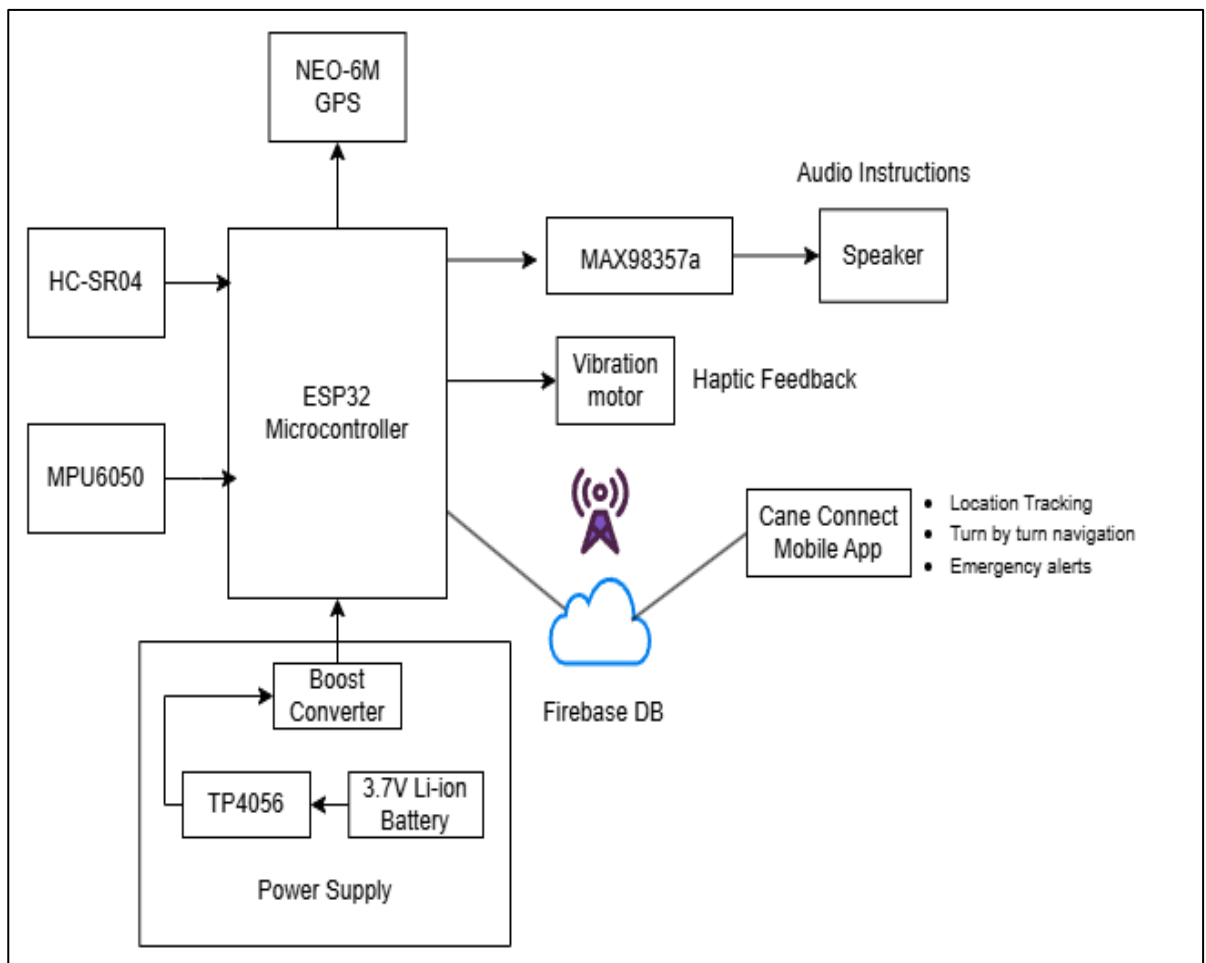


FIGURE 3.1: Block Diagram of the proposed system model.

3.2.1 Smart Cane Hardware

The Smart Cane is equipped with multiple sensors and modules which are meant for obstacle detection, a floor-fall detection system, and assisting with navigation. It has ultrasonic sensors for real-time obstacle detection to provide immediate haptic feedback to alert the user. The fall detection module uses an MPU6050 with 3-axis accelerometer and 3-axis gyroscope to continuously monitor motion patterns and detect sudden falls that automatically alert emergency contacts via Firebase. A GPS module is used for real-time location tracking, sending the information to the cloud for monitoring by caregivers. For navigation, dynamic instructions are generated using the Google Maps Directions API. Then the instructions are converted into audio files using the Android TTS API and sent to the cane, where it will be played through an I2S-based audio feedback system. The ESP32 microcontroller is the nucleus of the system that handles the sensor data processing, cloud communication, and real-time alert responses in order to maintain a frictionless and responsive assistive experience.

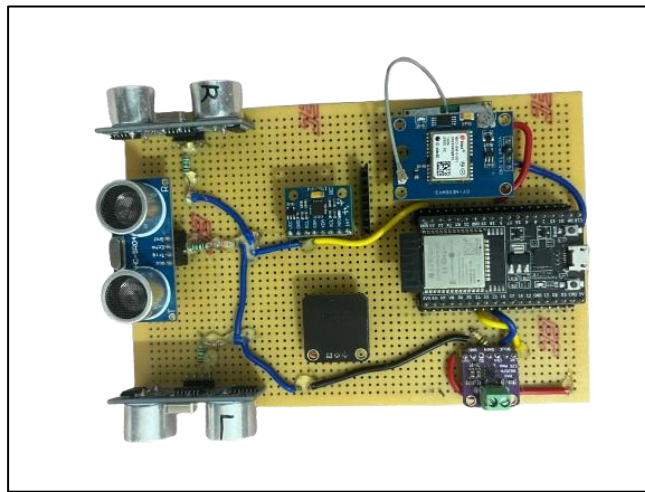


FIGURE 3.2: Proposed Smart Cane system integrated on a PCB.

3.2.2 Firebase Cloud Backend

Firebase real-time database and Firestore are the core functions to process and store data so as to create real-time communication between the Smart Cane and the mobile application. Firestore stores dynamic navigation instructions for the application. During navigation, instructions are navigated using the Google Maps Directions API and converted to speech files using Android TTS API before being saved in Firebase Storage. Links to these audio files are stored in Firestore. The Smart Cane retrieves them for referencing while navigating for response to real-time prompts. The onboard sensors continuously monitor for falls. The mobile application continually monitoring the changes to the flag will issue a local notification to inform the caregiver of the fall incident and current location of the user. The cane's GPS coordinates are also periodically updated in Firebase to allow real-time location tracking of the user by caregivers.

The fall detection feature is connected to Firebase Realtime Database for real-time updating. When fall happens, the system sets a flag "true," so it will automatically notice the caregiver with the user's location and fall condition. When the cane stabilizes, the flag will be reset to "false," notifying the caregiver about the recover. The caregiver is informed immediately and can respond quickly in emergencies. Therefore, reliability in synchronizing the hardware and the mobile application caters for an immediate response to emergency and security threats on the user's safety.

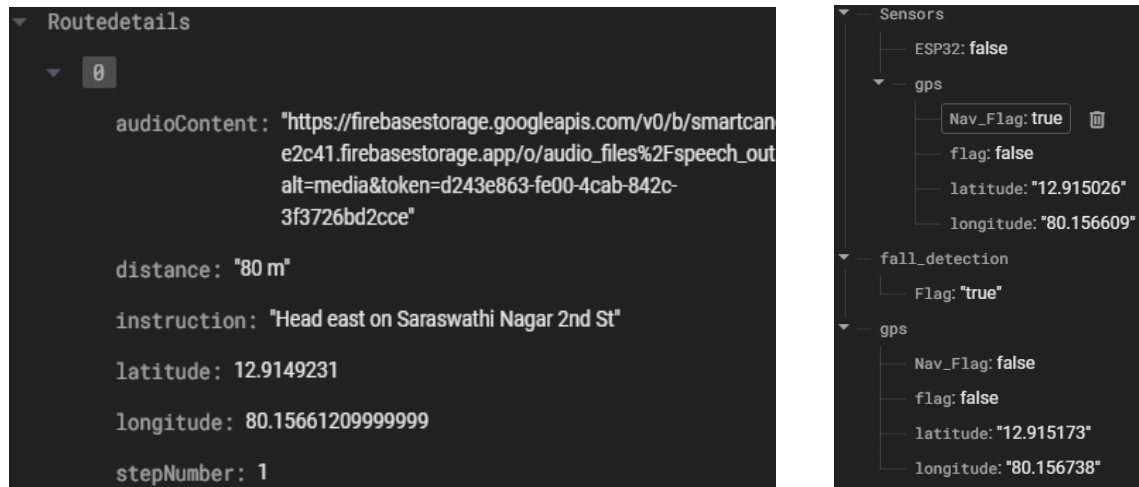


FIGURE 3.3: Structure of Firestore and Realtime Cloud Database

3.2.3 Cane Connect Mobile Application

The mobile application is the primary interface for caregivers to monitor the activity and provide real-time assistance to guarantee the safety of visually impaired users. The application is constantly fetching GPS coordinates using Firebase so that the user's current location can be accurately tracked on Google Maps. Caregivers can also input a destination into the app, which will then calculate a route and provide turn-by-turn instructions via the Google Directions API. The dynamic steps of navigation, together with distances and audio instructions, are stored in Firestore and are queried later by the Smart Cane for providing real-time voice navigation guidance. The application is updated through Firebase about any fall situation, thus notifying the caregiver about the user's location and fall status.

This app can still receive notifications even when the app is closed and running in the background. Subsequently, due to its impeccable sync with Firebase, the mobile application majorly aids in real-time tracking, navigation assistance, and fall detection, thereby augmenting the security and reliability of the Smart Cane system.

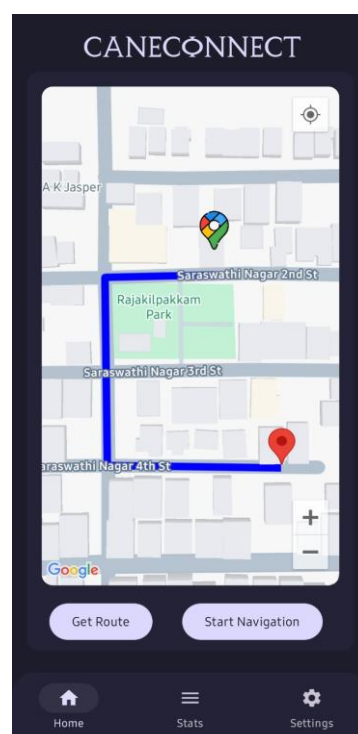


FIGURE 3.4: Home page of the Cane Connect mobile application

3.3 Hardware Implementation

3.3.1 Microcontroller Integration

The Smart Cane uses an ESP32 microcontroller as the main processing unit for controlling sensor input, navigation commands, and real-time data communication with Firebase. This microcontroller communicates with various hardware components through distinct communication protocols and GPIO pins. The direct interaction between the ESP32 microcontroller and other hardware components is listed below:

- i. **Obstacle Detection:** Three ultrasonic sensors are used for obstacle detection. They are connected to the ESP32 as follows:
 - Left Sensor: TRIG = GPIO 18, ECHO = GPIO 32
 - Straight Sensor: TRIG = GPIO 4, ECHO = GPIO 33
 - Right Sensor: TRIG = GPIO 5, ECHO = GPIO 21The ESP32 triggers each sensor sequentially and calculates the time taken for the ultrasonic pulse to return, determining the distance of obstacles.
- ii. **Fall Detection:** The MPU6050 accelerometer and gyroscope communicate via I2C and are connected to:
 - SDA = GPIO 19
 - SCL = GPIO 23The ESP32 continuously monitors acceleration and angular velocity to detect falls.
- iii. **GPS Tracking:** The NEO-6M GPS module is interfaced using UART with the following connections:
 - RX = GPIO 16
 - TX = GPIO 17The module provides real-time latitude and longitude data, which is sent to Firebase for tracking by caregivers.
- iv. **Audio Feedback System:** The MAX98357A I2S DAC module is connected via I2S for voice-based navigation instructions:
 - DOUT = GPIO 22
 - BCLK = GPIO 26
 - LRC = GPIO 25The ESP32 fetches pre-recorded navigation instructions from Firebase and plays them through the speaker using this module.
- v. **Vibration Motor:** A vibration motor is connected to GPIO 15, providing haptic feedback when obstacles are detected. The ESP32 controls the motor using PWM signals to modulate vibration intensity.
- vi. **Wireless Communication:** The ESP32's built-in Wi-Fi module handles data synchronization with Firebase, eliminating the need for additional communication modules.

An ESP32 module manages power through a 3.7V Li-ion battery, regulated by the TP4056 charging module; a 5V boost converter is incorporated to satisfy the higher voltage demands of the other components. The firmware is written in a way that allows the system to multitask by being in real-time and power-conscious.

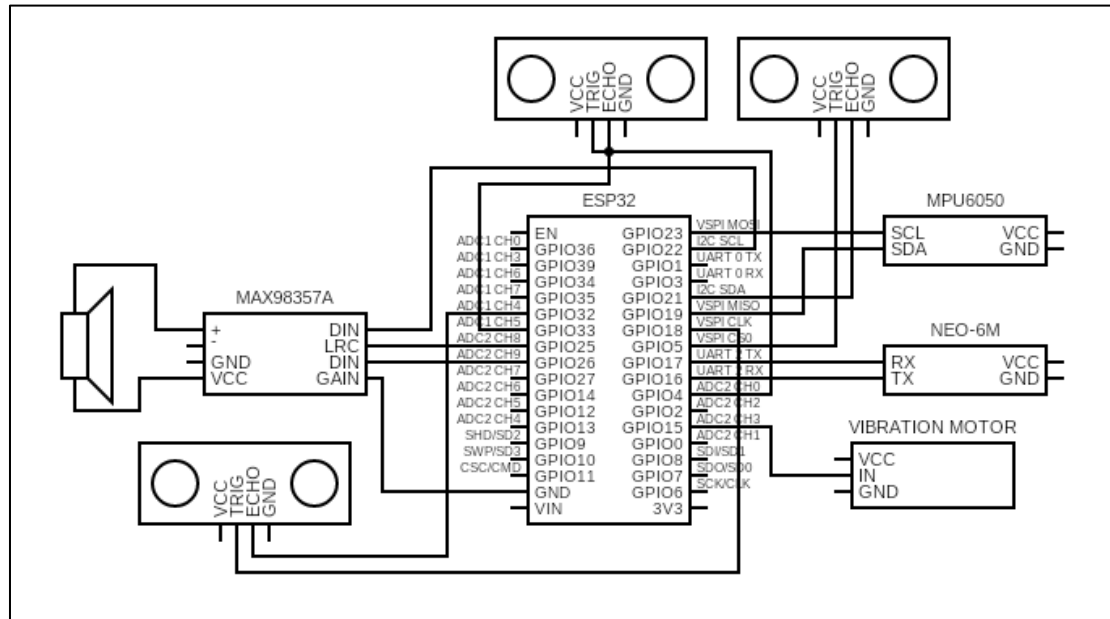


FIGURE 3.5: Connection diagram between ESP32 and other modules in the system

3.3.2 Obstacle Detection System

The smart cane's obstacle-detection system utilizes ultrasonic sensors to keep track of obstacles along the path of the user. This system is important for navigation safety by instantaneously providing feedback to the performer. The Cane consists of 3 ultrasonic sensors at different strategic points and angles along the cane, which continuously measure the distance to nearby obstacles. Feedback mechanism is incurred if there is an obstruction.

Sensor Placement and Connection:

To detect obstacles from various directions, three ultrasonic sensors are used:

- Left Sensor: Positioned on the left side of the cane to detect obstacles approaching from the left.
- Straight Sensor: Positioned in front of the cane to detect obstacles directly in the path of the user.
- Right Sensor: Positioned on the right side of the cane to detect obstacles coming from the right.

Each ultrasonic sensor has a trigger pin and echo pin. The trigger pin sends out an ultrasonic pulse, while the echo pin listens for when that pulse has bounced off an object and returned. The time period between the emission and reception of the pulse is proportional to the distance it has to travel to meet the object.

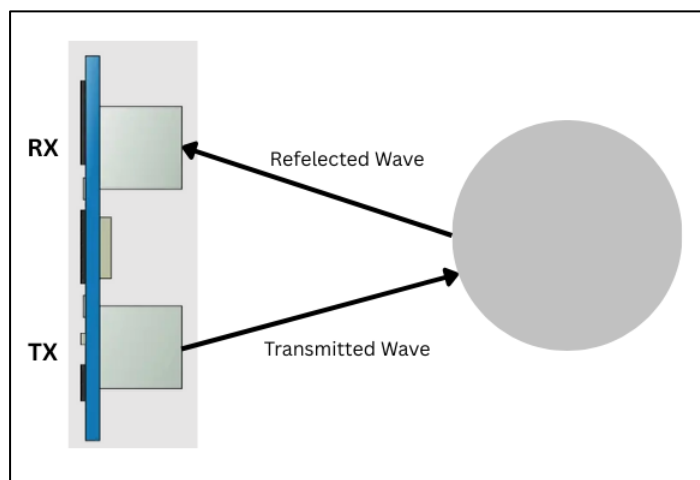


FIGURE 3.6: Working of HC-SRO4 sensor

The trigger and echo pins of each sensor are connected to the ESP32 microcontroller, which processes the sensor data and determines the distance to the detected object. The following pins on the ESP32 are used to interface with the sensors:

- Left Sensor: Trigger Pin (18) and Echo Pin (32)
- Straight Sensor: Trigger Pin (4) and Echo Pin (33)
- Right Sensor: Trigger Pin (5) and Echo Pin (21)

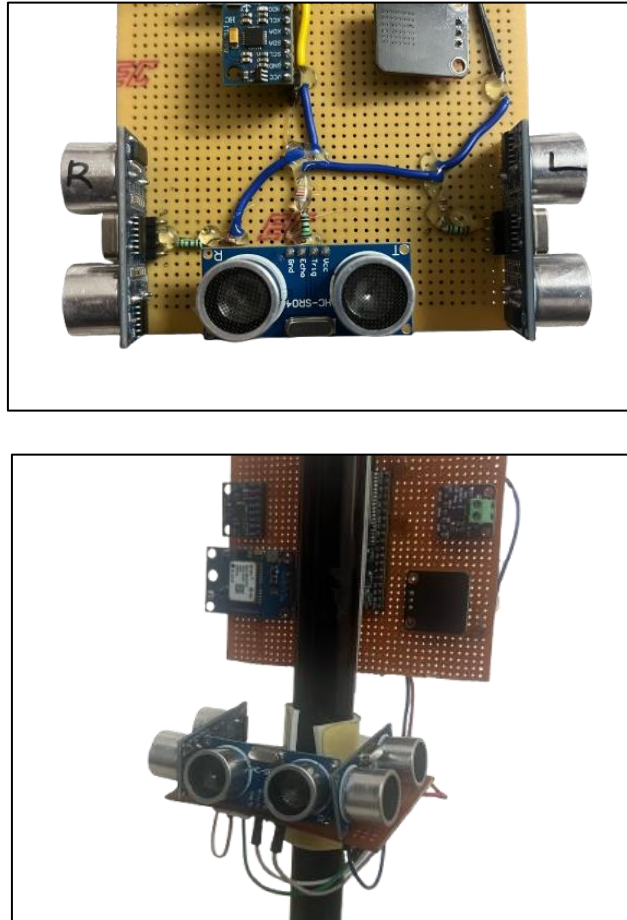


FIGURE 3.7: Placement of 3 ultrasonic sensors in the cane

Signal Processing and Detection Logic:

The obstacle detection system operates by emitting an ultrasonic pulse from the sensor's trigger pin. This pulse travels through the air, reflects off any nearby object, and returns to the sensor's echo pin. The time taken for the echo to return is measured, and this duration is used to calculate the distance to the object using the known speed of sound in air.

The speed of sound is approximately 343 meters per second, or 0.034 cm/ μ s. The distance is computed using the formula:

$$distance = \frac{duration \times 0.034}{2}$$

Here, duration represents the round-trip time in microseconds, and the division by 2 accounts for the two-way travel of the sound wave. If the calculated distance falls within a predefined threshold, the system confirms the presence of an obstacle and triggers the appropriate response, such as haptic feedback to the user.

To ensure accurate and reliable detection, the distance measurements from the ultrasonic sensors are thresholded within a defined operational range. The system only considers objects that fall within the following range:

- **Minimum Detection Distance (DETECTION_MIN):** 5 cm
- **Maximum Detection Distance (DETECTION_MAX):** 100 cm

Measurements under the minimum threshold or beyond the maximum threshold are also considered invalid and not processed, to avoid misdetection from sensor noise or environmental limitations. It is only when valid distance readings fall within this range that the haptic feedback system that flags the presence of nearby obstacles for the user is triggered.

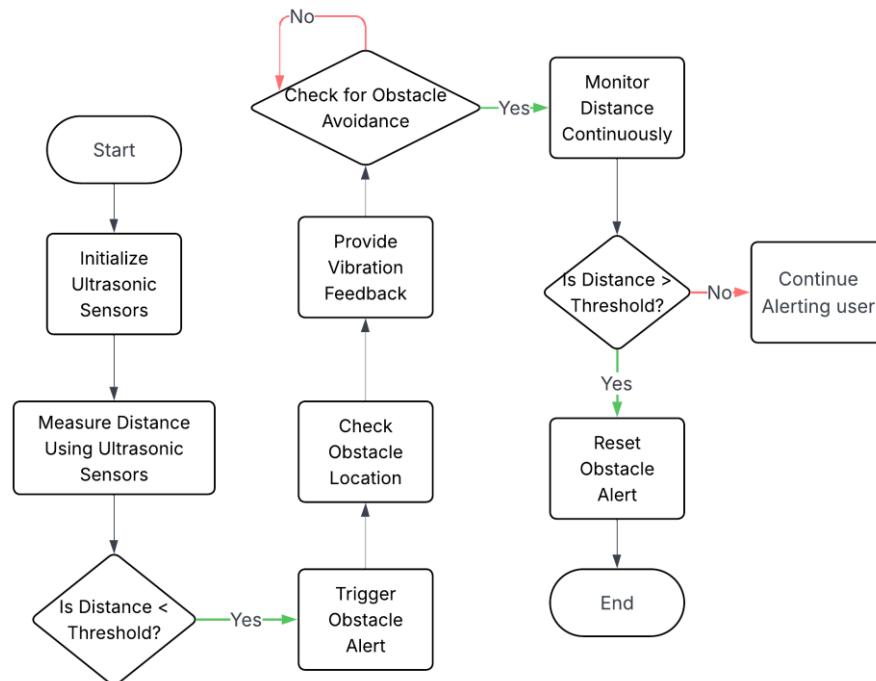


FIGURE 3.8: Flowchart illustrating obstacle detection and alert process.

Feedback Mechanism:

The user is warned by the system via the vibration motor once an obstacle is detected. The user gets haptic feedback about the presence of the obstruction through the vibration. The ESP32 sends a simple signal to activate the motor via its control of the VIB_PIN to create real vibration. The feedback continues for 500 milliseconds and then cuts off.

The distance is also checked in real-time for each sensor, meaning that a person will constantly give feedback when getting nearer to other obstacles.

3.3.3 Fall Detection System

The Fall Detection System of the Smart Cane utilizes the MPU6050 module, which consists of a 3-axis accelerometer and a 3-axis gyroscope. The accelerometer measures acceleration along the X, Y, and Z axes, while the gyroscope tracks angular velocity, allowing the system to monitor the cane's motion and orientation.

Sensor Integration:

The MPU6050 sensor measures the cane's acceleration and rotation in three axes (X, Y, Z). These readings help detect changes in the cane's position and motion. The system uses both accelerometer and gyroscope data to determine whether the cane has fallen and when it has been returned to an upright position.

Fall Detection Logic:

The fall detection system works based on the following logic:

1. **Stationary Cane:** When the cane is stationary and upright, the accelerometer values show Y-axis acceleration close to 9.8 m/s² (gravity), and X and Z values are low (below 1).
2. **Fall Detection:** A fall is detected when there is a significant change in acceleration (X or Z > 7.5, Y < 1), indicating a fall or sudden impact.
3. **Gyroscope Confirmation:** The gyroscope data confirms the fall if the gyro readings (gyroX, gyroY, gyroZ) are close to 0, indicating the cane is stationary.
4. **Recovery Detection:** After a fall, the system detects recovery when the Y-axis acceleration exceeds 7.5, and X and Z values are below 4, indicating the cane is upright again.

Firebase Integration:

When a fall is detected, the system updates the Firebase Realtime Database to set the fall status to "true," alerting the caregiver. Once recovery is detected, the fall status is reset to "false."

System Workflow:

1. The MPU6050 sensor continuously monitors the cane's movement.
2. Significant changes in the accelerometer or gyroscope data trigger a fall alert.
3. The system updates the fall status in Firebase and notifies the caregiver's mobile app.
4. The system detects recovery when the cane is upright and resets the fall status.
5. This system enhances user safety by ensuring that caregivers are immediately notified of falls and recoveries.

```
Acceleration - X: -5.10 Y: -3.53 Z: 6.85 | Gyroscope - X°: -0.05 Y°: 0.08 Z°: -0.04
Acceleration - X: -5.16 Y: -3.53 Z: 6.86 | Gyroscope - X°: -0.01 Y°: 0.07 Z°: -0.02
Acceleration - X: 0.43 Y: 0.48 Z: 9.24 | Gyroscope - X°: 0.09 Y°: 0.35 Z°: -0.03
Acceleration - X: 0.75 Y: -0.04 Z: 9.41 | Gyroscope - X°: 0.01 Y°: 0.05 Z°: -0.00
Acceleration - X: 0.74 Y: -0.06 Z: 9.37 | Gyroscope - X°: 0.01 Y°: 0.04 Z°: -0.00
Acceleration - X: -0.89 Y: -0.37 Z: 10.13 | Gyroscope - X°: -0.18 Y°: 0.49 Z°: 0.03
Acceleration - X: -7.65 Y: -1.81 Z: 5.21 | Gyroscope - X°: -0.09 Y°: 0.33 Z°: -0.07
Acceleration - X: -8.10 Y: -2.64 Z: 4.33 | Gyroscope - X°: 0.09 Y°: 0.18 Z°: -0.06
Acceleration - X: 2.15 Y: 0.39 Z: 1.95 | Gyroscope - X°: 0.02 Y°: 0.81 Z°: -0.20
Acceleration - X: 0.72 Y: -0.05 Z: 9.40 | Gyroscope - X°: 0.01 Y°: 0.06 Z°: 0.00
Acceleration - X: 0.71 Y: -0.05 Z: 9.41 | Gyroscope - X°: 0.02 Y°: 0.06 Z°: -0.00
Acceleration - X: 0.69 Y: -0.04 Z: 9.39 | Gyroscope - X°: 0.01 Y°: 0.05 Z°: -0.00
Acceleration - X: -0.02 Y: 0.22 Z: 10.20 | Gyroscope - X°: 0.32 Y°: 1.28 Z°: -0.09
Acceleration - X: -8.62 Y: 2.92 Z: -1.25 | Gyroscope - X°: -0.21 Y°: 0.02 Z°: 0.46
Acceleration - X: -8.90 Y: 3.66 Z: -0.02 | Gyroscope - X°: 0.44 Y°: 0.67 Z°: -0.56
Acceleration - X: -20.10 Y: 12.38 Z: 22.86 | Gyroscope - X°: 4.37 Y°: 1.60 Z°: 2.44
```

```
Acceleration - X: -3.05 Y: -0.36 Z: -9.51 | Gyroscope - X°: -0.13 Y°: -0.11 Z°: 0.01
High Acceleration - Possible Fall
Acceleration - X: 2.55 Y: -0.26 Z: -9.91 | Gyroscope - X°: 0.06 Y°: 0.68 Z°: -0.02
Fall Not Confirmed - Motion Detected
Updating Fall Status in Firebase... Success!
Acceleration - X: 3.13 Y: -0.48 Z: -9.89 | Gyroscope - X°: 0.02 Y°: 0.12 Z°: -0.01
High Acceleration - Possible Fall
Acceleration - X: 3.13 Y: -0.49 Z: -9.87 | Gyroscope - X°: 0.02 Y°: 0.05 Z°: -0.00
FALL CONFIRMED!
Updating Fall Status in Firebase... Success!
Acceleration - X: 3.15 Y: -0.48 Z: -9.85 | Gyroscope - X°: 0.02 Y°: 0.05 Z°: -0.00
FALL DETECTED
Cane is stationary after fall
Acceleration - X: 3.13 Y: -0.49 Z: -9.85 | Gyroscope - X°: 0.02 Y°: 0.06 Z°: -0.00
FALL DETECTED
Acceleration - X: 3.12 Y: -0.47 Z: -9.86 | Gyroscope - X°: 0.02 Y°: 0.06 Z°: -0.01
FALL DETECTED
Acceleration - X: 3.12 Y: -0.46 Z: -9.85 | Gyroscope - X°: 0.02 Y°: 0.06 Z°: -0.00
FALL DETECTED
Acceleration - X: 3.13 Y: -0.47 Z: -9.87 | Gyroscope - X°: 0.02 Y°: 0.06 Z°: -0.00
FALL DETECTED
Acceleration - X: 3.12 Y: -0.49 Z: -9.89 | Gyroscope - X°: 0.02 Y°: 0.06 Z°: -0.01
FALL DETECTED
```

FIGURE 3.9: Serial Monitor output for the Fall Detection system

3.3.4 Turn-by-Turn Navigation

The turn-by-turn navigation system of Smart Cane is an audio-based guidance system in real time using input from GPS, Firebase, and speech playback to allow users to navigate their path to a particular destination. The NEO-6M GPS module is used together with a MAX98357A DAC for audio output in land navigation purposes.

Hardware Components:

The NEO-6M GPS module continuously provides location data to the ESP32, which processes the coordinates, comparing them to GPS coordinates for the next waypoint. Once the cane is within the vicinity of a waypoint, while the following audio instructions are being played from Firebase Storage as per real-time guidance through the speaker, they are converted from Firebase Storage into sound via the MAX98357A DAC. Playback of audio instructions through the I2S interface of the ESP32 thus makes the audio instructions clear and unambiguous-such instructions could be, "Turn left," or "Continue straight."

Navigation Logic:

The system computes the distance to the next waypoint based on GPS data. Once a user is within 10 meters of their next step, an audio instruction for that step will be triggered. The whole procedure is repeated until the user reaches the final destination. Route details including audio file links are stored in Firestore, while navigation status is tracked through the Nav_Flag in Firebase Realtime Database.

3.3.5 Audio Feedback

The Smart Cane system employs an Audio Feedback System to provide real-time, turn-by-turn auditory navigation for visually impaired users. The backbone of the system is the MAX98357A I2S Digital-to-Analog Converter (DAC) module interfaced directly with the ESP32 microcontroller through the I2S protocol. This module will receive digital audio data streaming from the ESP32, and convert it into analog signals. On top of decoding, the MAX98357A comes with an integrated Class-D audio amplifier, which amplifies the output signal to drive an 8Ω, 0.5W speaker, making sure that the playback is clear and audible outdoors.

The system dynamically fetches pre-recorded audio instruction files saved in Firebase Storage that are linked to the different navigation steps in the RouteDetails structure in Firebase Firestore. Therefore, as the user traverses the route, the system GPS identifies the current waypoint and retrieves the appropriate audio instruction in real time. Together with the MAX98357A DAC for signal processing and amplification, and seamless integration with Firebase services and the ESP32, Smart Cane is able to deliver precise, timely, and intelligible audio feedback for each navigation cue as assistance to its users.

3.3.6 Power Management System

The Smart Cane runs on rechargeable lithium-ion battery with a voltage of 3.7V, selected for its size and efficiency. The battery connects to the TP4056 charging module, which is responsible for securely charging the battery when connected to an appropriate power supply. The TP4056 module facilitates the battery's efficient charging while protecting it from overcharging and deep discharge, thus contributing to battery life.

The components of the Smart Cane are powered by stepping up the voltage with a boost converter. The boost converter steps up a battery voltage of 3.7V to 5V required by various components, such as the ESP32 microcontroller, some ultrasonic sensors, and the MAX98357A DAC module. This configuration makes sure the Smart Cane can work effectively without requiring several batteries, maintaining a design that is simple and provides adequate power to run the system.

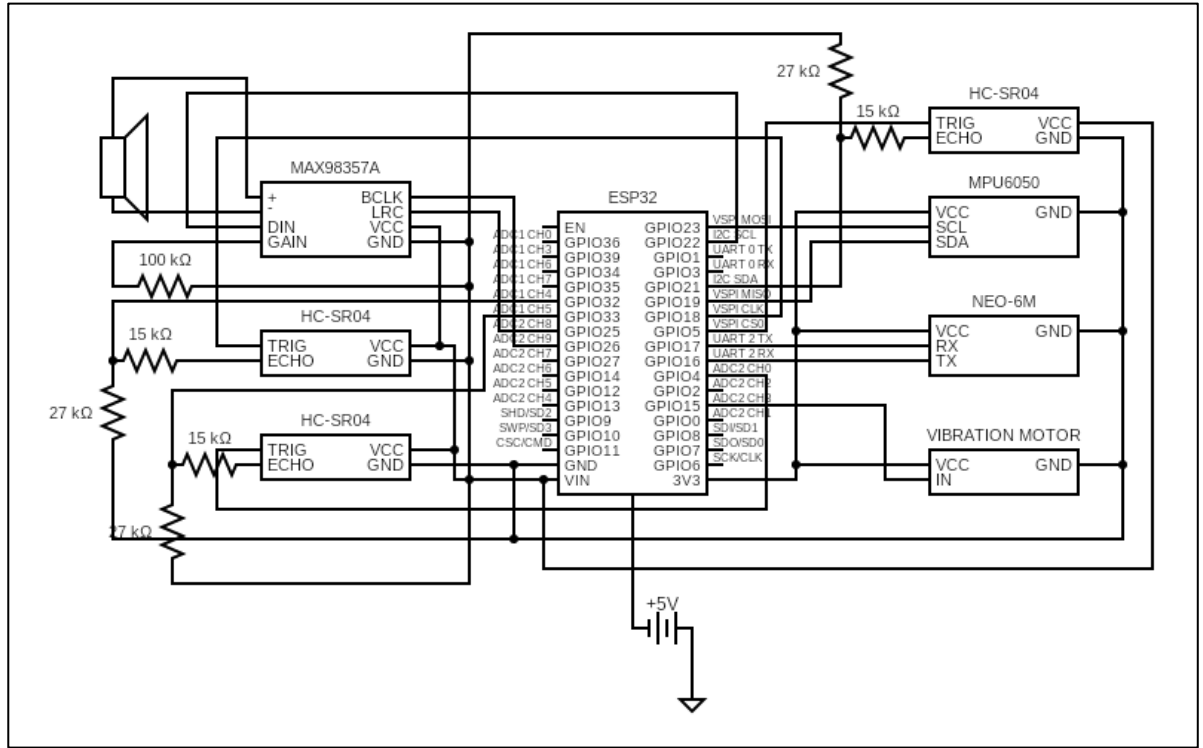


FIGURE 3.10: Circuit Diagram of the proposed system

3.4 Software Implementation

3.4.1 Cane Connect Mobile Application

The Cane Connect mobile application serves as the central interface between the caregiver and the Smart Cane system. It is developed using Android's modern development tools and frameworks, including Jetpack Compose, ViewModel, Kotlin Coroutines, Firebase, and the Google Maps SDK. The app integrates multiple modules to enable real-time tracking, fall detection alerts, and turn-by-turn navigation feedback.

1. Real-Time Location Tracking

The application continuously listens to the Smart Cane's GPS location updates via Firebase Realtime Database. The cane's latitude and longitude are stored in the database, and the app reflects these updates in real time using a LocationViewModel with a StateFlow-based structure. The user's location is plotted on the map using the Google Maps Compose API, with support for toggling map types and dynamically updating markers or polylines.

2. Fall Detection Notifications

The app uses a background service (FallDetectionService) that runs as a foreground service, ensuring persistent monitoring even when the app is closed or the phone restarts. It listens for changes in the fall detection status. If a fall is detected, it immediately triggers a high-priority local notification to alert the caregiver. This service is auto-started on device boot using a BootReceiver.

3. Turn-by-Turn Navigation Integration

When the caregiver selects a destination, the app fetches navigation instructions using the Google Directions API via Retrofit. The request returns a polyline that is decoded using PolyUtil.decode() and displayed as a visual path on the map. Each step in the directions is parsed into a list of NavigationStep objects, which includes:

- Step number

- Text instruction
- Distance
- GPS coordinates
- Audio content

These steps are then uploaded to Firestore under the Users/user001/Routedetails field. This structure allows the ESP32 on the Smart Cane to fetch and follow each step sequentially.

4. Audio Navigation Feedback for the User

The parsed step instructions are stored in a `MutableStateList` and are intended to be converted to audio either via Text-to-Speech (TTS) or pre-recorded audio files. These are played through a MAX98357A I2S DAC module attached to the cane's ESP32 controller. This enables the visually impaired user to receive real-time spoken directions while walking.

5. Firestore Communication and Data Handling

Firestore is used as the main interface for step-wise navigation details, while Firebase Realtime Database is used for fast, low-latency updates such as location tracking and fall detection status. The app uses `viewModelScope.launch` blocks with `await()` and `awaitResponse()` to ensure asynchronous and efficient communication with both Firestore and external APIs.

6. State Management and Modularity

The app follows modern Android architecture principles by using:

- `ViewModel` for business logic separation.
- `mutableStateOf`, `mutableStateListOf`, and `StateFlow` for UI reactivity.
- Coroutines for background tasks and API calls.

A modular and scalable codebase structure supporting clear interaction between UI, `ViewModel`, and backend logic.

Technologies Used:

1. **Google Maps API:** Displays the Smart Cane's real-time location and navigation path on the map.
2. **Google Directions API:** Generates step-by-step navigation routes from the user's current location to the destination.
3. **Firebase Realtime Database:** Stores and syncs real-time GPS data, fall detection status, and system flags between the cane and the app.
4. **Firebase Firestore:** Manages structured navigation data including route steps, instructions, and associated audio file links.
5. **Firebase Storage:** Hosts audio files for navigation instructions, which are retrieved by the cane for playback.
6. **Retrofit:** Handles HTTP requests for APIs like Google Directions, ensuring efficient data communication.
7. **Local Notifications:** Sends high-priority alerts to caregivers when a fall is detected.
8. **Foreground Service:** Maintains continuous background operations such as fall monitoring and location updates.
9. **Boot Receiver:** Automatically restarts essential services after a device reboot to ensure uninterrupted functionality.
10. **SharedPreferences:** Stores user-specific settings like alert preferences and navigation configurations.
11. **ViewModel:** Preserves UI-related data and ensures smooth screen updates during state changes.
12. **NavController:** Manages seamless screen navigation within the app.

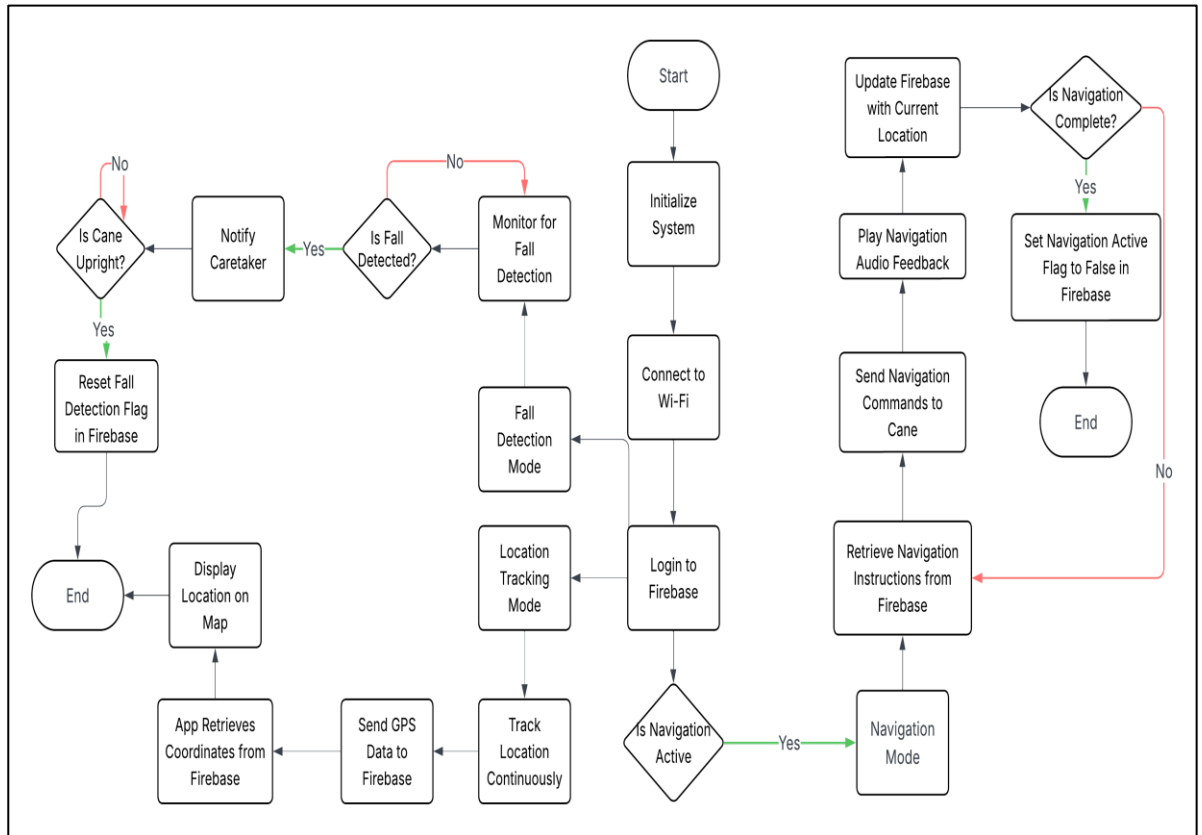


FIGURE 3.11: Flowchart illustrating the working of the mobile application

3.4.2 GPS and Location tracking

The Smart Cane relies on merging the NEO-6M GPS module with the ESP32 for tracking real-time geographic coordinates of the user. The latitude and longitude data are parsed and processed by an efficient GPS library called TinyGPSPlus. The GPS coordinates are transmitted, uninterruptedly, to the Firebase Realtime Database for accurate and up-to-date tracking of location.

On the other hand, in the mobile-app front, the app listens to real-time updates from Firebase. The map tracks the cane's position dynamically as the cane moves by getting the updated coordinates from the Firebase server. This enables the caregiver to continuously monitor the user for live tracking and location-based assistance whenever required.

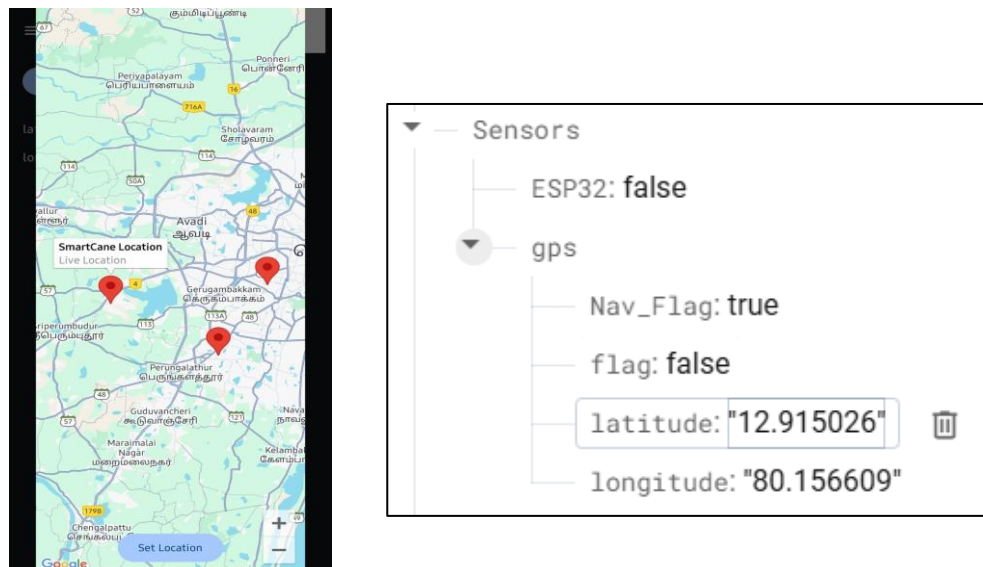


FIGURE 3.12: Live location tracking and RTDB structure

Function to fetch GPS data and update Firebase every 10 seconds:

```

void loop() {
  while (gpsSerial.available() > 0) {
    gps.encode(gpsSerial.read());
  }

  if (gps.location.isValid()) {
    float lat = gps.location.lat();
    float lon = gps.location.lng();

    if (Firebase.ready()) {
      unsigned long currentMillis = millis();
      if (currentMillis - sendDataPrevMillis > 10000) {
        sendDataPrevMillis = currentMillis;
        updateRTDB(lat, lon);
      }
    }
  }
}

void updateRTDB(float lat, float lon) {
  FirebaseJson json;
  json.set("/latitude", String(lat, 6));
  json.set("/longitude", String(lon, 6));
  if (Firebase.RTDB.updateNode(&fbdo, gpsDatabasePath.c_str(), &json)) {
    Serial.println("RTDB updated (GPS)");
  }
}

```

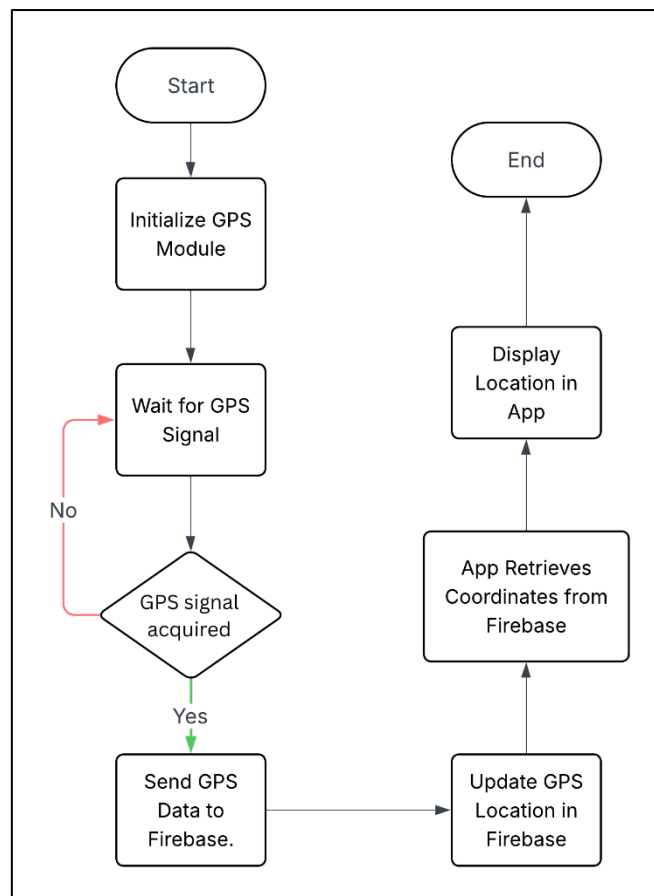


FIGURE 3.13: Flowchart illustrating live location tracking process.

3.4.3 Fall Detection

The fall detection system is implemented by using the MPU6050 sensor together with the ESP32 that is interfaced to perform real-time monitoring of motion and orientation data. It continuously analyzes the acceleration and gyroscope values to detect sudden changes that could indicate a fall. Once such a detect pattern is established, the system confirms that both the cane is in a non-upright position and that it remains still to confirm the fall. If confirmed, it updates a fallDetected flag in the Firebase Realtime Database with respect to whether a fall had occurred.

The mobile application then listens continuously to changes in that Firebase flag through a Foreground Service. Fall detection triggers an immediate local notification of high priority for caregivers, whether the app runs in the foreground, in the background, or even after a reboot (this is handled by BootReceiver). Timely notifications are thus ensured for emergencies, with the notification carrying important fall status details assigned as well as real-time updates for caregivers overseeing the current status of the user and responding promptly.

Fall Detection Algorithm:

The code initializes the MPU6050 accelerometer and gyroscope sensors using the Adafruit_MPU6050 library. This provides accelerometer (a) and gyroscope (g) data for fall detection.

Fall Detection States: The fall detection algorithm works in a series of states (FallState enum):

- i. **NO_FALL:** The cane is in a stable position. If the acceleration on the X or Z axis exceeds a threshold (indicating a fall), the state transitions to FALL_DETECTED.
- ii. **FALL_DETECTED:** Potential fall detected based on the acceleration data. The gyroscope is checked for minimal motion ($\text{gyroX}, \text{gyroY}, \text{gyroZ} < 0.1$). If the gyroscope values indicate the cane is stationary, the fall is confirmed, and the state transitions to FALL_CONFIRMED.
- iii. **FALL_CONFIRMED:** A fall has been confirmed based on both acceleration and gyroscope data. The system waits for recovery, transitioning to FALL_RECOVERY once the cane is stationary after the fall.
- iv. **FALL_RECOVERY:** The recovery state checks if the cane is upright again (acceleration values in a stable range). Once recovery is detected (cane picked up), the state resets back to NO_FALL.

Data Collection:

The loop continuously checks sensor data and applies the fall detection logic based on the current fallState. If a fall is detected and confirmed, it triggers relevant actions, like updating the fall status in Firebase and notifying the caretaker.

Stability and Recovery Check:

Recovery is detected if the cane returns to an upright position, which is monitored through the accelerometer's Y-axis value (with a threshold of >7.5 indicating the cane is upright). If this happens, the fall state resets, indicating recovery.

Function for Fall Detection:

```
void loop() {
  switch (fallState) {
    case NO_FALL:
      if ((accelX > 7.5 || accelZ > 7.5) && accelY < 4) {
        Serial.println("High Acceleration - Possible Fall");
        fallState = FALL_DETECTED;
      }
    
```

```

}
  break;

case FALL_DETECTED:
  if (gyroX < 0.1 && gyroY < 0.1 && gyroZ < 0.1) {
    Serial.println("FALL CONFIRMED!");
    fallState = FALL_CONFIRMED;
    fall = true;
  } else {
    fallState = NO_FALL;
  }
  break;

case FALL_CONFIRMED:
  Serial.println("Cane is stationary after fall");
  fallState = FALL_RECOVERY;
  break;

case FALL_RECOVERY:
  if (accelY > 7.5 && accelX < 4 && accelZ < 4) {
    Serial.println("Recovery detected - Cane picked up");
    fallState = NO_FALL;
    fall = false;
  }
  break;
}
}

```

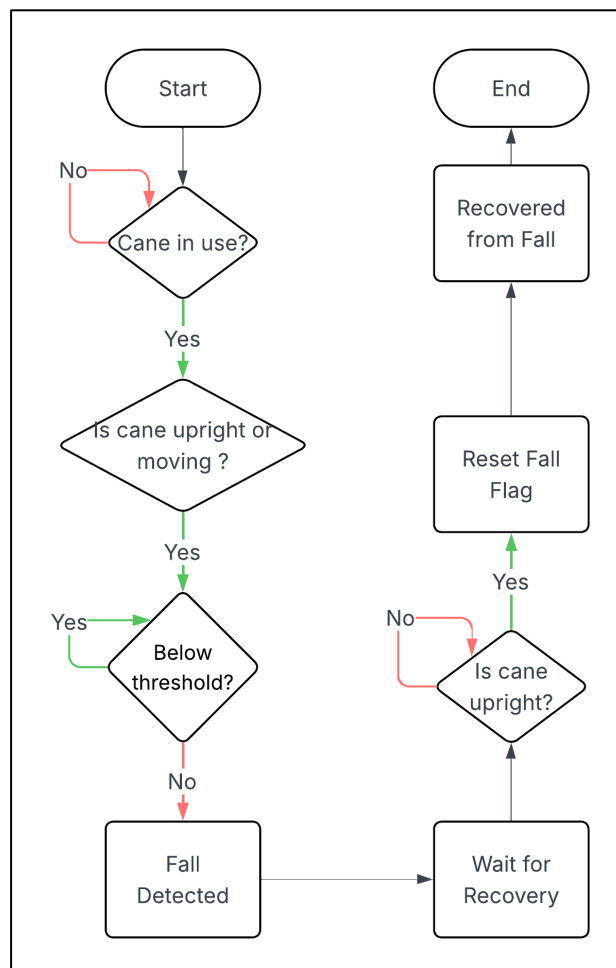


FIGURE 3.14: Flowchart illustrating fall detection process.

3.4.4 Turn-by-Turn navigation

The Smart Cane provides turn-by-turn navigation to allow assisted movement with audio instructions. In the ESP32 system, the GPS continuously gets the position of the cane, and as the user approaches a navigation waypoint, the next instruction is retrieved from the Firebase Firestore. Each navigation step includes the target latitude, longitude, and an audio file link that points to Firebase Storage. This audio content is played through an I2S DAC (MAX98357A) to a speaker, thus clearly giving the turn-by-turn guidance to the user. After the final instruction step, the system sets a navigationActive = false flag in Firebase to signify that the route is completed.

At the mobile application end, when the caregiver selects a destination, the application creates a route using the Google Directions API, starting at the cane's GPS location. This route is broken down into stepwise directions, each with text-based instructions and respective coordinates. These steps, along with the associated links to the audio files, are uploaded to Firestore under that user's document. This architecture allows the cane to independently fetch and follow instructions for seamless, location-aware audio navigation.

Function for turn-by-turn navigation implementation:

```
bool playInstruction(int stepNumber) {
    if (stepNumber > totalSteps || stepNumber < 1 || audioUrls[stepNumber - 1] == "") {
        return false;
    }
    audio.stopSong();
    int retries = 2;
    while (retries > 0 && !audio.connecttohost(audioUrls[stepNumber - 1].c_str())) {
        delay(500);
        audio.stopSong();
        retries--;
    }
    if (retries > 0) {
        while (audio.isRunning()) {
            audio.loop();
            delay(1);
        }
        return true;
    }
    return false;
}

void fetchNavFlag() {
    if (Firebase.ready()) {
        if (Firebase.RTDB.getBool(&fbdo, "/Sensors/gps/Nav_Flag")) {
            Nav_Flag = fbdo.boolData();
        }
    }
}

void loop() {
    if (Firebase.ready() && (millis() - sendDataPrevMillis > timerDelay || sendDataPrevMillis == 0)) {
        sendDataPrevMillis = millis();
        updateRTDB(lat, lon);
    }
    fetchNavFlag();

    if (Nav_Flag && currentStep <= totalSteps) {
        if (playInstruction(currentStep)) {
```



```

currentStep++;
if (currentStep > totalSteps) {
    Nav_Flag = false;
    audio.stopSong();
}
}

```

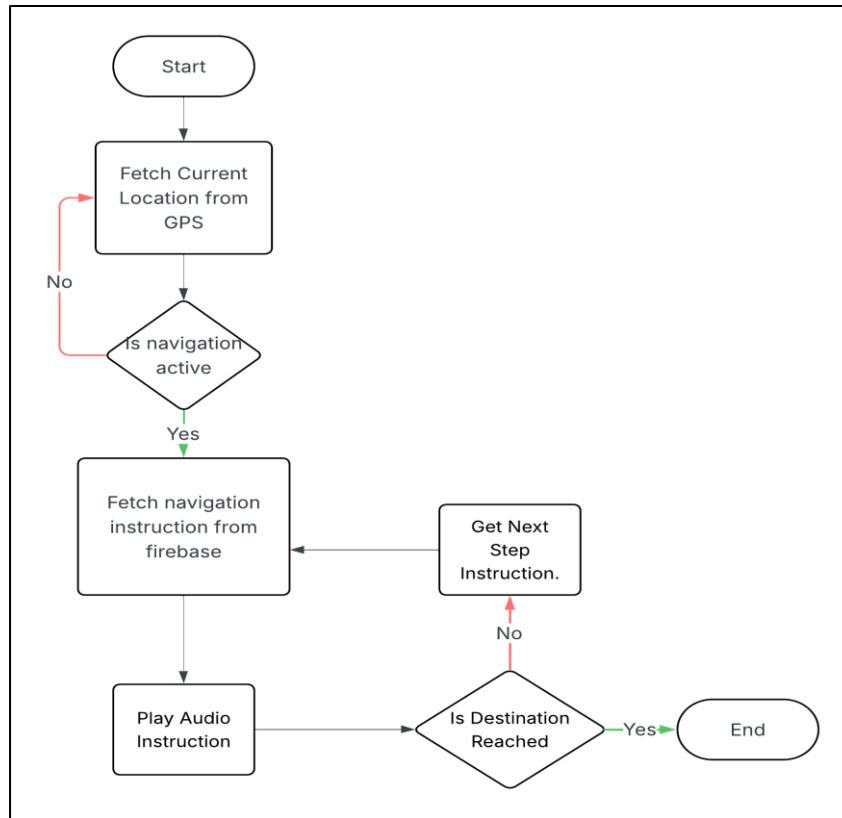


FIGURE 3.15: Flowchart illustrating turn-by-turn navigation process.

3.4.5 Firebase Integration

Firebase is crucial for enabling real-time communication and synchronization between the smart cane and the caregiver's mobile application. It acts as the data transport layer for fall alerts, live location sharing, and navigation coordination. The main highlights of firebase integration in the system include:

1. **Firebase Realtime Database:** The real-time database for Firebase is used to store dynamic data such as GPS readings from the Smart Cane, fall-detection status, and navigation activity status. It provides real-time data updates that are immediately reflected in the caregiver's app.
2. **Firestore Database:** Used to store structured data in a generalized way for doing turn-by-turn navigation, with coordinates, directions, and audio file links associated with each step. During navigation, route data can thus be retrieved with ease.
3. **Firebase Storage:** Stores the pre-recorded audio files for the guidance of navigation. These audio files are fetched by the ESP32 and played through the DAC module to guide the user.
4. **Real-Time Listeners:** The smart cane and mobile app are equipped with listeners to detect changes in the database. For example, when a fall is detected and a flag is set, the app instantly gets this update and triggers a notification.
5. **Two-way Communication:** Firebase permits this two-way data flow. The app uploads navigation routes to Firestore while the cane reads and reacts to them. Similarly, the cane pushes live updates (location/fall status) displayed by the app in real time.

Thus, Firebase provides a seamless, low-latency medium of communication between hardware and software components, thus making the Smart Cane responsive, connected, and friendly.

Firestore Initialization:

The Firestore configuration is initialized at the beginning of the software setup to establish a connection between the ESP32 and the Firestore platform. The API key, database URL, and authentication credentials are provided to set up the communication between the Smart Cane and Firestore. The ESP32 uses Firestore Authentication to securely log in using email and password credentials.

Code for firestore initialization:

```
FirestoreConfig config;
FirestoreAuth auth;
FirestoreData fbdo;

config.api_key = API_KEY;
auth.user.email = USER_EMAIL;
auth.user.password = USER_PASSWORD;
config.database_url = DATABASE_URL;
Firestore.begin(&config, &auth);
```

Real-Time Data Upload and Retrieval:

Smart Cane sends GPS coordinates and navigation updates (Nav_Flag) to Firestore Realtime Database regularly. It also gets back the navigation flag and fall detection so as to determine the navigation mode and trigger alerts. This real-time streaming of data guarantees that both the device and mobile app have the updated information.

Code for data upload to RTDB:

```
void updateRTDB(float lat, float lon) {
    FirestoreJson json;
    json.set("/latitude", String(lat, 6));
    json.set("/longitude", String(lon, 6));
    json.set("/Nav_Flag", Nav_Flag);
    Firestore.RTDB.updateNode(&fbdo, gpsDatabasePath.c_str(), &json);
}
```

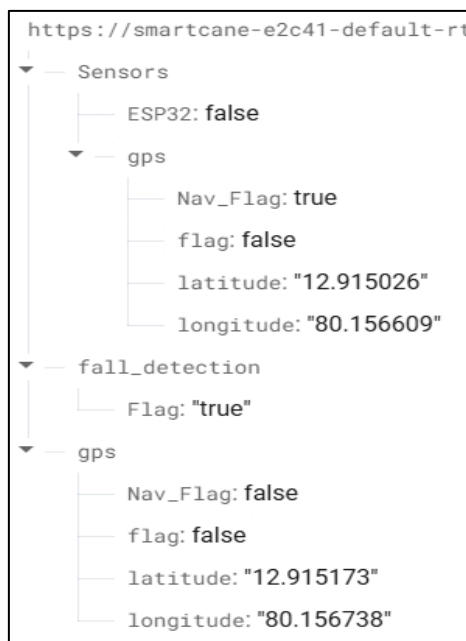


FIGURE 3.16: Data upload to RTDB in real-time.

Fetching Navigation Instructions from Firestore

Firestore stores navigation instructions in the form of detailed audio instructions. The Smart Cane retrieves these instructions on the navigation route, offering the user audio guidance. Firebase Firestore is used to save detailed information on audio files and route steps for seamless navigation experience.

Code to fetch navigation instructions:

```
void checkFirestoreData() {
    HTTPClient http;
    String url = "https://firestore.googleapis.com/v1/projects/" +
String(FIREBASE_PROJECT_ID) + "/databases/(default)/documents/Users/user001";
    http.begin(url);
    int httpCode = http.GET();
    if (httpCode == HTTP_CODE_OK) {
        String jsonPayload = http.getString();
        StaticJsonDocument<2048> doc;
        DeserializationError error = deserializeJson(doc, jsonPayload);
        if (!error) {
            totalSteps = doc["fields"]["Routedetails"]["arrayValue"]["values"].size();
            for (int i = 0; i < totalSteps && i < 6; i++) {
                audioUrls[i] =
doc["fields"]["Routedetails"]["arrayValue"]["values"][i]["mapValue"]["fields"]["audioCont
ent"]["stringValue"].as<String>();
            }
        } else {
            totalSteps = 0;
        }
    } else {
        totalSteps = 0;
    }
    http.end();
}
```

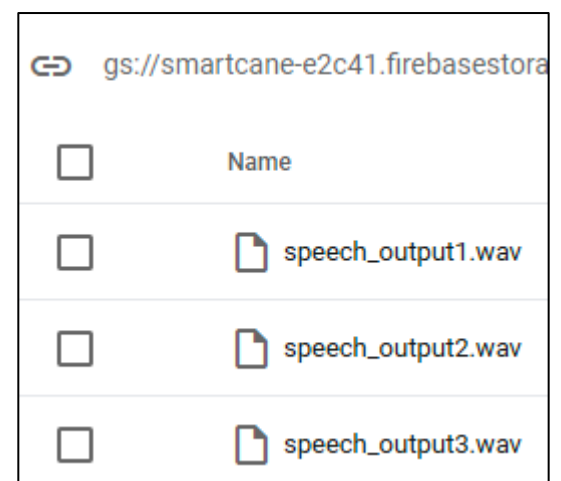
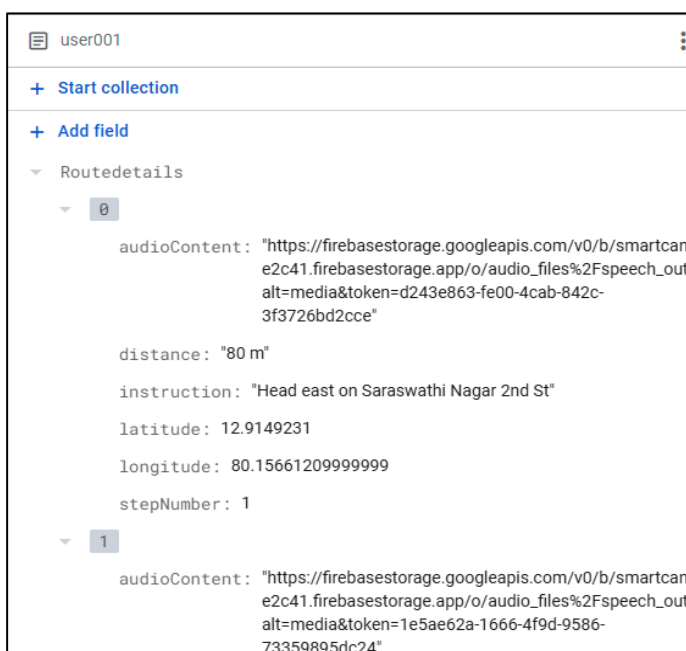


FIGURE 3.17: Navigation instructions and audio saved in Firestore and Firebase data storage.

Firestore Database Structure:

The Smart Cane makes use of Firestore for storing the steps involved in the navigation. The RouteDetails spans a document structure within the Users collection, and comprises step-wise information about the navigation route along with the distance, audio instructions, and coordination.

The Firestore Database structure for storing navigation instructions is given below:

- **Collection:** Users
 - **Document:** user001
 - **Array Field:** RouteDetails
 - **Each Step:** Contains:
 - i. **audioContent:** Firebase Storage link to the audio file for the step's instructions
 - ii. **instruction:** Text-based instruction for the navigation step
 - iii. **distance:** Distance to the next step
 - iv. **latitude:** Latitude for the step's location
 - v. **longitude:** Longitude for the step's location
 - vi. **stepNumber:** Sequence number of the step in the navigation route

Each time the user sets a destination on the mobile app, Firebase generates step-by-step navigation instructions and stores them in Firestore. The Smart Cane fetches this data to play the audio instructions in real-time to guide the user. This structure ensures that navigation instructions are stored in a scalable way, with each step being easily accessible for playback and navigation feedback.

```

"user001": {
  "Routedetails": [
    {
      "audioContent": "https://firebasestorage.googleapis.com/v0/b/smartcane-e2c41.firebaseio.com/o/audio",
      "distance": "80 m",
      "instruction": "Head east on Saraswathi Nagar 2nd St",
      "latitude": 12.9149231,
      "longitude": 80.15661209999999,
      "stepNumber": 1
    },
    {
      "audioContent": "https://firebasestorage.googleapis.com/v0/b/smartcane-e2c41.firebaseio.com/o/audio",
      "distance": "0.2 km",
      "instruction": "Turn right at Global OrchidPass by Manju akka house (on the left)",
      "latitude": 12.9149179,
      "longitude": 80.1573492,
      "stepNumber": 2
    },
    {
      "audioContent": "https://firebasestorage.googleapis.com/v0/b/smartcane-e2c41.firebaseio.com/o/audio",
      "distance": "0.1 km",
      "instruction": "Turn left at Palamlati Nivas onto Saravana Nagar 2nd StDestination will be on the left",
      "latitude": 12.9128855,
      "longitude": 80.1573903,
      "stepNumber": 3
    }
  ],
  "gyro_status": false
}

```

FIGURE 3.18: Firestore DB structure.

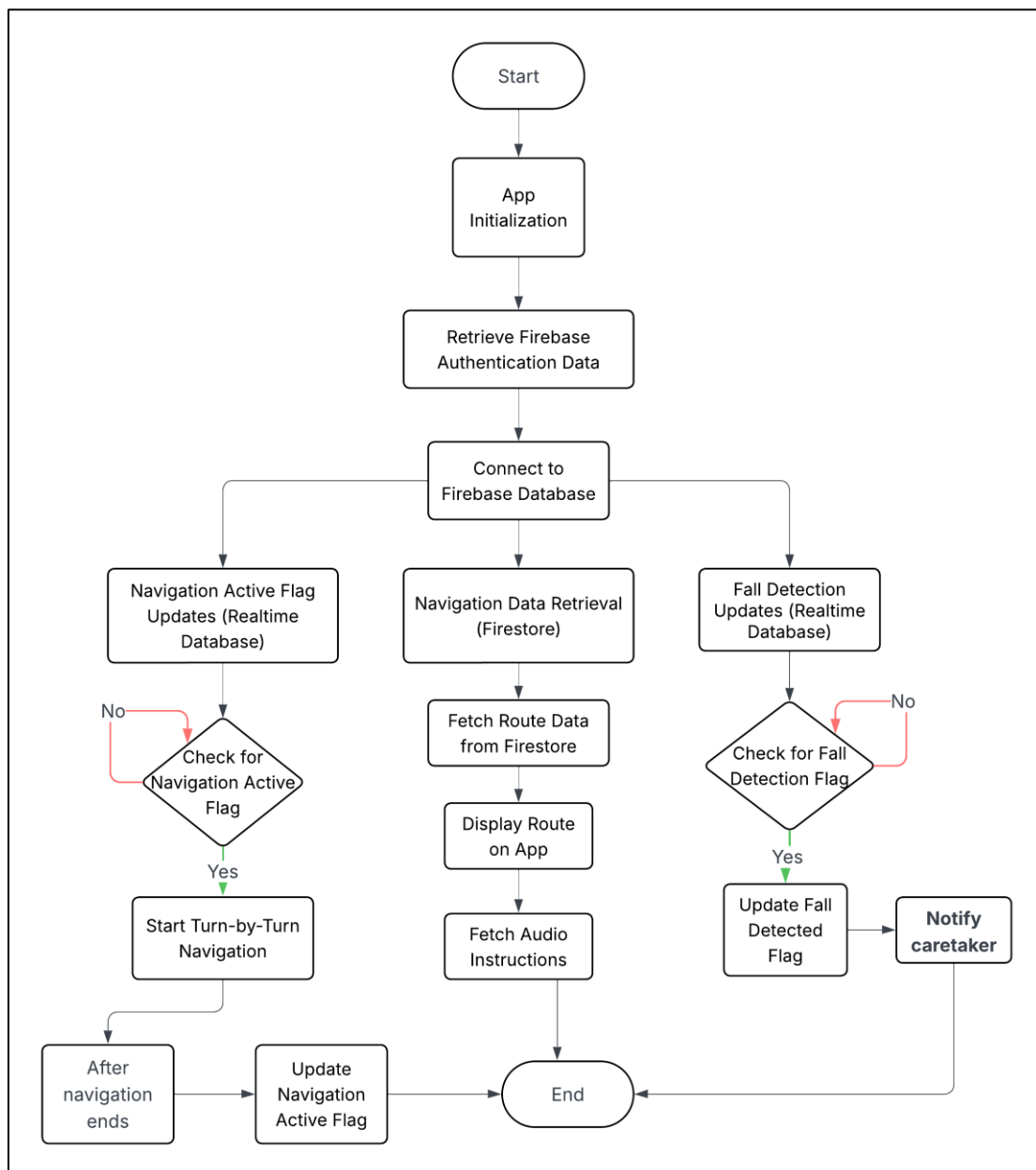


FIGURE 3.19: Flowchart illustrating the working of Firebase.

3.5 Conclusion

In conclusion, the Smart Cane project involves a methodology which integrates hardware as well as software components to further enhance fall detection capabilities, navigation, and obstacle avoidance. Real-time data is gathered using the sensors MPU6050 and ultrasonic, while data are stored efficiently via Firebase with real-time updates and communication to the caregivers. The combination of GPS provides location tracking and turn-by-turn navigation so that the Smart Cane can serve both safety and mobility assistance, making this Smart Cane a complete solution for the elderly and visually impaired.

Chapter 4

Results and Discussions

Objective

The primary objective of the Smart Cane was to create a multifunctional, fully specialized assistive device for the benefit of visually impaired users regarding safety and independence. The project planned to provide a system that monitored falls, tracked the location of the user in real-time, navigated turn-by-turn, and detected obstacles on the user's path.

The Smart Cane operates by interlinking the sensors of the Smart Cane with the mobile app for constant monitoring and interaction. The system is built to detect a fall and sends out an immediate alert to the caregivers so they can respond promptly when in need. Real-time location tracking allows caregivers to know where the user is at all times. The navigation system takes the user step-by-step and provides immediate audio-visual feedback to the user, thus making navigation in an unfamiliar locality much easier and safer. Obstacle detection mainly enables users to bypass obstacles at their immediate destinations, thereby boosting their mobility.

Developing an integration between the cane and mobile application creates an efficient platform for data exchange, ensuring real-time communication for updates, notifications, and navigation. The goal was to establish a robust, easy-to-use, and inclusive kind of system that allows visually impaired persons to move freely and confidently while providing extra comfort for caretakers via timely updates and alerts.

4.1 System Performance and Integration

The Smart Cane system was designed and implemented to assist the visually impaired with real-time navigation assistance, fall detection, and obstacle avoidance. The system performed as expected during testing, with each feature achieving the desired functionality:

1. **Fall Detection:** The fall detection system worked well, detecting falls based on the accelerometer and gyroscope readings, where the cane correctly identified stationary and dynamic states. It could differentiate between falling and normal activity, thereby minimizing false positives. In real-time, the Firebase cloud database would be updated to notify the caretaker on the reception of a fall detection call.

```

Acceleration - X: -0.01 Y: -9.79 Z: -0.48 | Gyroscope - X°: 0.02 Y°: 0.05 Z°: -0.01
Acceleration - X: 1.21 Y: -10.05 Z: -1.07 | Gyroscope - X°: 0.05 Y°: 0.32 Z°: 0.05
Acceleration - X: -5.21 Y: -2.63 Z: -10.31 | Gyroscope - X°: -0.42 Y°: 0.18 Z°: 0.32
High Acceleration - Possible Fall
Acceleration - X: -1.74 Y: 0.22 Z: -10.68 | Gyroscope - X°: -0.01 Y°: -0.07 Z°: -0.40
Fall Not Confirmed - Motion Detected
Updating Fall Status in Firebase... Success!
Acceleration - X: -5.20 Y: -0.90 Z: -8.76 | Gyroscope - X°: 0.07 Y°: -0.20 Z°: 0.00
High Acceleration - Possible Fall
Acceleration - X: -3.90 Y: -0.72 Z: -9.59 | Gyroscope - X°: -0.06 Y°: 0.33 Z°: 0.26
Fall Not Confirmed - Motion Detected
Updating Fall Status in Firebase... Success!
Acceleration - X: -6.15 Y: -0.06 Z: -7.67 | Gyroscope - X°: 0.07 Y°: 0.84 Z°: 0.26
High Acceleration - Possible Fall
Acceleration - X: -3.88 Y: -1.42 Z: -8.92 | Gyroscope - X°: -0.07 Y°: 0.03 Z°: -0.27
Fall Not Confirmed - Motion Detected
Updating Fall Status in Firebase... Success!
Acceleration - X: -5.77 Y: -0.74 Z: -8.54 | Gyroscope - X°: 0.16 Y°: -0.20 Z°: 0.08
High Acceleration - Possible Fall
Acceleration - X: -2.51 Y: -1.37 Z: -9.64 | Gyroscope - X°: -0.05 Y°: 0.00 Z°: -0.09
FALL CONFIRMED!
Updating Fall Status in Firebase... Success!
Acceleration - X: -5.60 Y: -0.48 Z: -8.60 | Gyroscope - X°: 0.09 Y°: -0.10 Z°: 0.05
FALL DETECTED

```

```

FALL DETECTED
Acceleration - X: 0.62 Y: -9.80 Z: -0.90 | Gyroscope
FALL DETECTED
Recovery detected - Cane picked up
Updating Fall Status in Firebase... Success!
Acceleration - X: 0.04 Y: -9.85 Z: -0.62 | Gyroscope
Acceleration - X: 0.03 Y: -9.82 Z: -0.64 | Gyroscope
Acceleration - X: 0.25 Y: -9.82 Z: -0.56 | Gyroscope

```

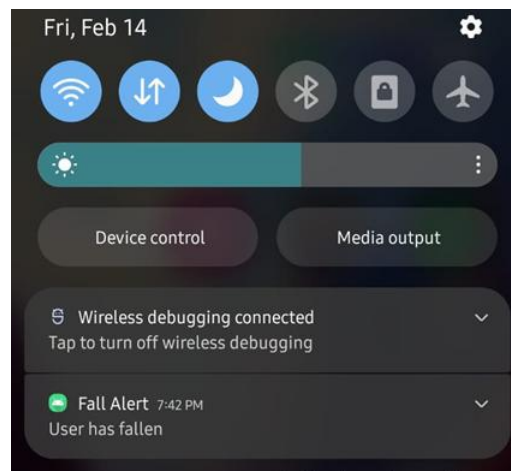


FIGURE 4.1: Serial monitor output for fall detection and notification.

2. **Location Tracking:** Using the NEO-6M GPS module, GPS coordinates were captured and sent to the mobile application. Real-time location updates allowed for the monitoring of the user via the Google Maps interface by the caretaker. The system demonstrates reliability even in moderately strong GPS conditions, providing accurate positioning with minimal latency.

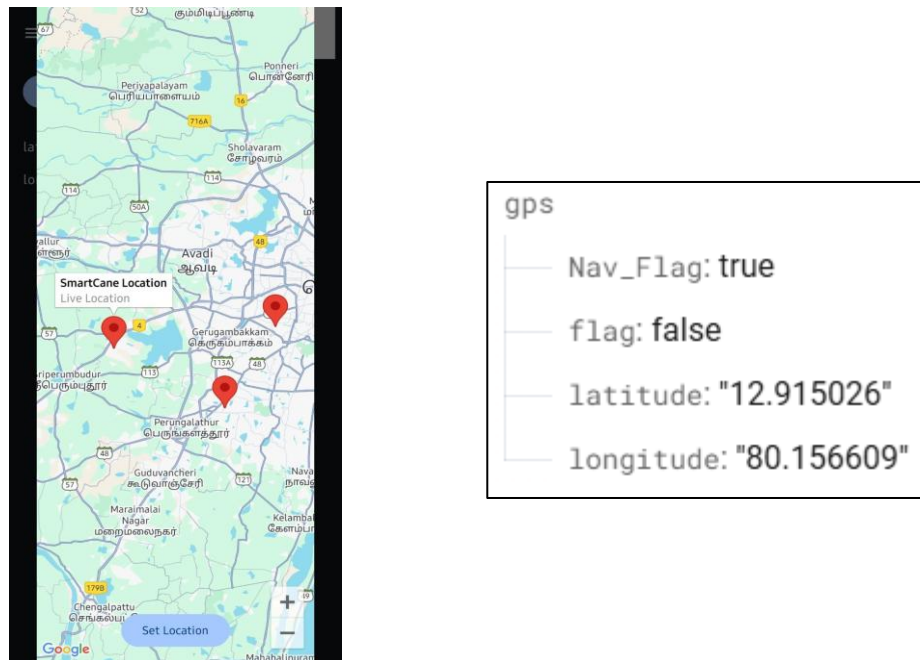


FIGURE 4.2: Live location tracking using mobile app and GPS coordinates uploaded to RTDB.

3. **Navigation Feedback:** With the clear audio feedback provided for every turn, the user will be able to easily follow the prescribed routing. In addition, the navigation steps could be fetched successfully from Firebase Firestore, with the playback of audio files through the I2S DAC system. The voice instructions were issued in a clear and timely manner throughout the navigation process. Also, the mobile interface showed the route on the map, thereby keeping navigation expectations aligned for both the user and the caretaker.

```

{
  audioContent: "https://firebasestorage.googleapis.com/v0/b/smartcane2c41.firebaseio.com/o/audio_files%2Fspeech_out%2Falt=media&token=d243e863-fe00-4cab-842c-3f3726bd2cce"
  distance: "80 m"
  instruction: "Head east on Saraswathi Nagar 2nd St"
  latitude: 12.9149231
  longitude: 80.15661209999999
  stepNumber: 1
}

```

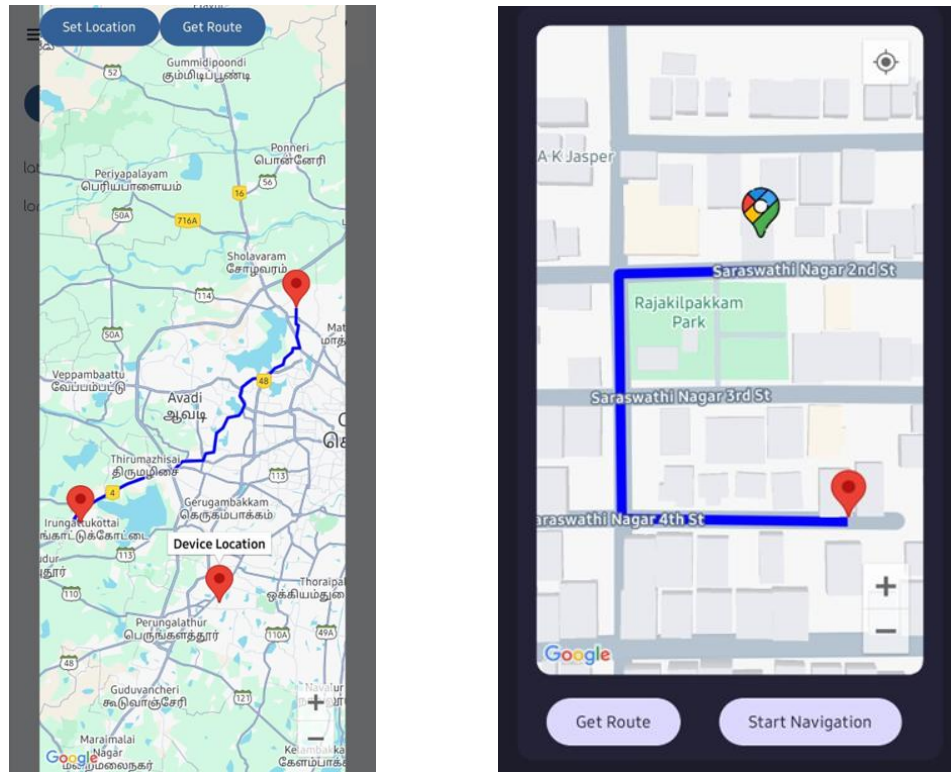
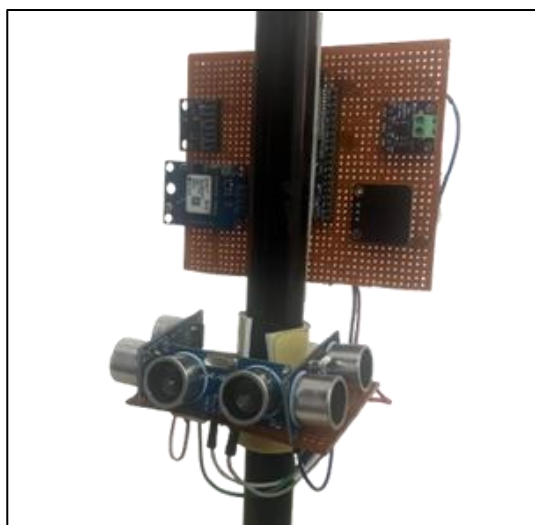



FIGURE 4.3: Navigation instructions and route calculation.

4. **Obstacle Detection:** The ultrasonic sensors were successfully detecting obstacles in the way of cane, and with real-time feedback mechanism, alerts were triggered when ever objects were detected within a given range. The system provided user feedback that guided users away from obstacles affording their mobility, thus making the cane more useful in changing environments.



```

Left: -1 cm, Straight: -1 cm, Right: -1 cm
Left: -1 cm, Straight: -1 cm, Right: -1 cm
Left: -1 cm, Straight: -1 cm, Right: -1 cm
Left: -1 cm, Straight: 19 cm, Right: 99 cm
Left: -1 cm, Straight: -1 cm, Right: -1 cm
Left: 12 cm, Straight: -1 cm, Right: 7 cm
Left: -1 cm, Straight: -1 cm, Right: -1 cm
Left: 12 cm, Straight: -1 cm, Right: -1 cm
Left: 11 cm, Straight: 12 cm, Right: -1 cm
Left: 14 cm, Straight: 7 cm, Right: -1 cm
Left: 7 cm, Straight: 8 cm, Right: -1 cm
Left: 11 cm, Straight: -1 cm, Right: -1 cm
Left: -1 cm, Straight: -1 cm, Right: -1 cm
Left: -1 cm, Straight: -1 cm, Right: -1 cm
Left: -1 cm, Straight: -1 cm, Right: -1 cm
Left: -1 cm, Straight: -1 cm, Right: -1 cm
Left: -1 cm, Straight: 12 cm, Right: -1 cm
Left: -1 cm, Straight: 16 cm, Right: -1 cm
Left: 11 cm, Straight: -1 cm, Right: -1 cm
Left: -1 cm, Straight: -1 cm, Right: -1 cm
Left: 8 cm, Straight: -1 cm, Right: -1 cm
Left: -1 cm, Straight: -1 cm, Right: -1 cm

```

FIGURE 4.4: Ultrasonic sensor placement and serial monitor output for obstacle detection.

4.2 Integration and Communication:

The integration of various subsystems and components formed an integral part of the Smart Cane project. The successful communication between mobile application, IoT hardware, and Firebase database assured that the entire system worked as a single unit.

1. **Mobile App and IoT Integration:** The mobile application acted as the main interface between the user, caretaker, and the Smart Cane system. It is mainly responsible for getting the real-time updates from the IoT system such as the user's GPS location, falling detection status, and navigation direction updates. Communication between the app and Smart Cane built with ESP32 was seamless and data was transmitted over Wi-Fi. The app was also capable of sending commands to the Smart Cane for navigation purposes, which the hardware would then interpret and use to guide the user.

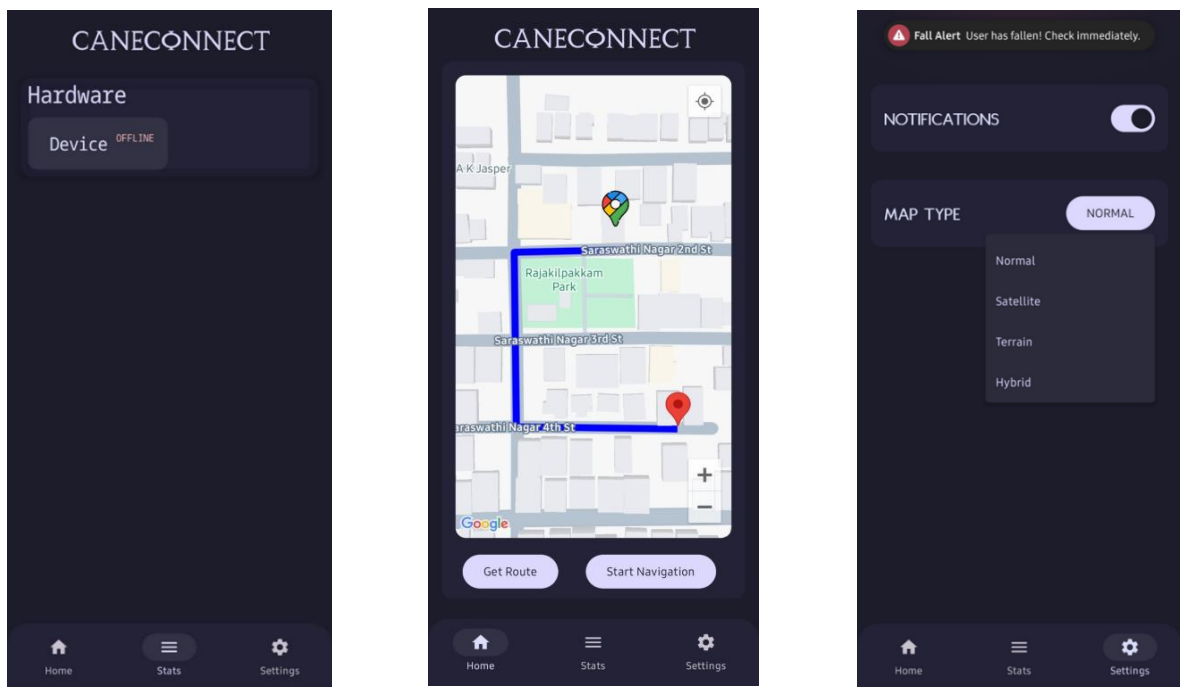


FIGURE 4.5: Cane Connect mobile app interface.

2. **Firestore as a Central Hub:** Firestore provided vitally needed communication in bridging the app and IoT hardware. Flags like `fallDetected` and `navigationActive` were updated in the real-time database environment, while Firestore took care of storing the navigation instructions in the background. As Firestore is a cloud service, it enabled constant app updates so that users and caretakers can easily access any updates on the Smart Cane status and navigation steps.
3. **Real-Time Data Flow:** The mobile application, Smart Cane, exchanged real-time data reliably and transfer effortlessly to the Firebase database. It kept updating fall alerts, navigation steps, and user location immediately, which ensured notification to caretakers once a fall was marked, and effective guidance provided to the users while navigating.

Prototype Integration and Assembly:

The prototype of the Smart Cane integrated different components, and all the essential parts were housed on a zero PCB board. This zero PCB board gave flexibility in manual arrangement and compact organization of the modules to bring about seamless communication among the microcontroller, sensors, and power supply. The microcontroller (ESP32) was placed centrally on the PCB, while the accelerometer, gyroscope, and ultrasonic sensors were carefully placed around it. Once components were in place, each of them was manually soldered for accuracy to ensure stable connections, avoiding problems such as short circuit or weak connections. Furthermore, the power management circuitry that included Li-Ion battery and a boost converter was soldered on this PCB for proper distribution to power the system.

Along with placing the prototype in the protective enclosure, the complete system will comprise a protective solution for housing all the components in a manner that protects them from the impacts of damage and environmental influences. The whole assembly can then be attached to any of the canes for fully operational and ready-to-use smart cane once the casing is put in place. It further ensures that the system remains completely functional and protected during its use by rendering it robust and effective in terms of durability and resilience.

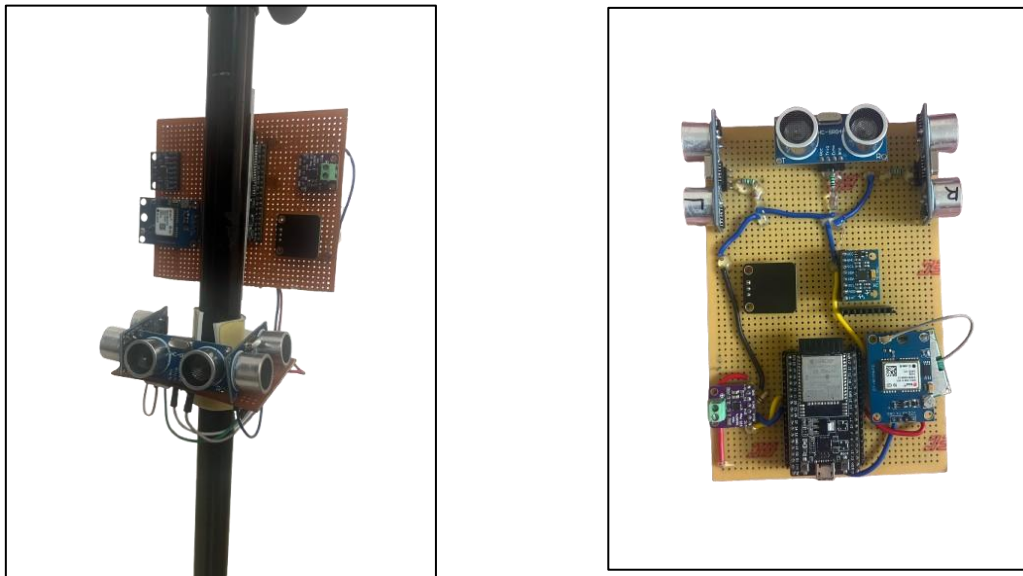


FIGURE 4.6: Prototype assembly of proposed Smart cane.

4.3 Accuracy and Reliability

The accuracy and reliability of Smart Cane, hence are considered core to the operations of Smart Cane in a real-world perspective. For fall detection, we adjusted the accelerometer and gyroscope sensors to measure parameters indicative of an actual fall. The system used the combined data from sensors, including the accelerometer- X, Y, and Z values and the gyroscope- angular velocities to decisive information regarding whether the cane had actually fallen. The fall detection algorithm was tested in different situations so that false positives (indicating a fall incorrectly) and false negatives (not sensing a fall) could be kept to the minimum possible.

The accuracy of GPS was tested by comparing the positions of the system with known coordinates in a controlled environment. Also, the turn-by-turn navigation feedback was tested for the clarity and promptness of the audio cue given to the user to follow the path with the least possible confusion.

The reliability of all components of the system was tested by subjecting them to stress tests to ensure that the system would continuously operate without failure. The system was subjected to tests for smooth wireless communication with smart cane and real-time alerts via Firebase. The system also underwent tests in various environments; indoors and outdoors were just two of them to ensure it would work in various settings. In summary, the smart cane system generally exhibited high accuracy and reliability such that the fall detection and navigation functions would perform as expected even under typical usage conditions.

4.4 Prototype Testing and Feedback

The Smart Cane model underwent elaborate testing toward evaluating its performance reliability and usability. Testing was done to assess the performance of major features like fall detection, obstacle detection, and navigation. There were quite a few tests to see if fall detection was actually sensitive to artificial falls, ensuring reliable feedback of trip obstacle detection within different environments, as well as verifying that the turn-by-turn navigation would give proper instructions and audio back.

The system functioned as it should, but there were a couple of areas that needed improvement. The volume of the speaker could be elevated, ensuring the audio feedback is loud enough for users in noisy environments. Besides that, and once a protective casing has been designed and attached, it will also serve the purpose of making the system mountable, securely installed into any cane for better portability and usability of the device. These are some changes scheduled for future releases to improve the whole experience for the user even more.

4.5 Challenges and Limitations

1. **Limited Memory on ESP32 Board:** The ESP32's limited memory caused memory allocation issues, especially when processing audio files for turn-by-turn navigation. This limitation impacted the ability to manage resources efficiently.
2. **GPS Module Performance:** The GPS module took a long time to acquire a satellite fix, and its accuracy was inconsistent, which occasionally led to unreliable location data. The module also only worked effectively outdoors.
3. **Low Audio Output Volume:** The 0.5-watt speaker used in the prototype had low volume output, making it difficult for the user to hear audio feedback in noisy environments. Additionally, the Android Text-to-Speech (TTS) system did not amplify the audio instructions, resulting in further volume limitations.
4. **Latency Issues:** When all modules were operating together, there was a noticeable delay in playing audio feedback. This high latency led to a disjointed user experience, particularly during turn-by-turn navigation.

5. **Limited Range of Ultrasonic Sensors:** The ultrasonic sensors had a relatively short detection range, which restricted the cane's ability to identify obstacles at a distance. This limitation could affect the cane's effectiveness in certain scenarios.
6. **System Performance Degradation:** Integrating multiple modules occasionally resulted in performance drops or delays due to resource conflicts. Each module required processing power and communication bandwidth, which sometimes led to slower operation.
7. **Obstacle Detection Limitations:** The obstacle detection system struggled to detect steps and uneven surfaces, which are common in daily environments. As a result, the cane might not be able to reliably warn users about obstacles like stairs or curbs.
8. **Environmental Sensitivity:** The current obstacle detection system was not capable of detecting certain environmental hazards such as water, fire, or other potentially dangerous conditions. This limited the system's ability to alert the user about environmental risks.

Chapter 5

Standards

5.1 IEEE 802.11 (Wi-Fi)

IEEE 802.11 refers to a set of standards for wireless local area networking (WLAN). These define the methods for wireless communication over 2.4 GHz and 5 GHz frequency bands. The ESP32 in the Smart Cane uses the 802.11b/g/n standards to connect to the internet and exchange data with Firebase for real-time services like fall detection, location updates, and navigation.

5.2 NMEA 0183

NMEA 0183 is a communication protocol developed for marine and GPS devices. It specifies the electrical signal, data format, and transmission method. In this project, the NEO-6M GPS module transmits location data to the ESP32 using the NMEA 0183 format via UART.

5.3 UART (Universal Asynchronous Receiver-Transmitter)

UART is a serial communication protocol used to transfer data between two devices without a clock signal. It operates using a transmit (TX) and receive (RX) line. The Smart Cane uses UART to receive GPS data from the NEO-6M module.

5.4 I2C (Inter-Integrated Circuit)

I2C is a two-wire serial protocol used to connect low-speed peripherals to microcontrollers. It supports multiple masters and slaves on the same bus. The MPU6050 motion sensor uses the I2C protocol to send acceleration and gyroscope data to the ESP32.

5.5 I2S (Inter-IC Sound)

I2S is a serial interface protocol for transmitting digital audio signals between devices. In this project, the ESP32 uses I2S to send audio instructions to the MAX98357A DAC module, which outputs sound for turn-by-turn navigation.

5.6 USB

USB is a standardized interface used for both communication and power supply between electronic devices. In this project, USB is utilized for programming the ESP32 and for providing regulated charging and power delivery to the system components, following the USB 2.0 standard to ensure reliable connectivity and energy efficiency.

5.7 HTTPS

HTTPS is a secure communication protocol used over the internet. It ensures encrypted data transmission. Both the ESP32 and mobile application use HTTPS to securely interact with Firebase and other cloud services.

Chapter 6

Conclusion and Future Scope

The Smart Cane project achieves an important integration of essential assistive technologies—fall detection, turn-by-turn navigation, and obstacle avoidance. It sends notifications to caregivers and provides navigational feedback for the user. Thus, it is a step towards ensuring safety and independence for visually impaired persons. Despite obstacles like memory constraints and GPS inaccuracies, one can say that the system performance illustrates the feasibility of integrating diverse IoT and sensor technologies. This furthers the need for innovation to provide options that enhance the accessibility and quality of life for persons with visual impairment.

There are numerous improvement and enrichment areas to pursue. The most important improvement will be related to the memory limitation of the ESP32 and then enhancing the audio for the user. The GPS system will need to be upgraded in terms of accuracy, using either other sensors or machine learning techniques. Another area of expansion is the field of obstacle detection because it does not yet recognize types of obstacles such as steps, water, or fire. The major improvement is going to be in the design and integration of a protective case which will be prototyped by 3D-printing and can be easily attached to any cane. Future work relates to exploring whether more advanced technologies, such as deep learning for object recognition, could be integrated to improve navigation and safety. Such enhancements will address existing issues and enable the Smart Cane to be a dependable as well as flexible tool for visually challenged users.

Chapter 7

Appendix

```
void initWiFi() {
    WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
    Serial.print("Connecting to WiFi");
    while (WiFi.status() != WL_CONNECTED) {
        Serial.print('.');
        delay(1000);
    }
    Serial.println("\nWiFi connected");
}

void checkFirestoreData() {
    HTTPClient http;
    String url = "https://firestore.googleapis.com/v1/projects/" +
String(FIREBASE_PROJECT_ID) +
"/databases/(default)/documents/Users/user001";
    http.begin(url);
    int httpCode = http.GET();
    if (httpCode == HTTP_CODE_OK) {
        String jsonPayload = http.getString();
        StaticJsonDocument<2048> doc;
        DeserializationError error = deserializeJson(doc, jsonPayload);
        if (!error) {
            totalSteps = doc["fields"]["Routedetails"]["arrayValue"]["values"].size();
            for (int i = 0; i < totalSteps && i < 6; i++) {
                audioUrls[i] =
doc["fields"]["Routedetails"]["arrayValue"]["values"][i]["mapValue"]["fields"
][["audioContent"]["stringValue"].as<String>());
            }
        } else {
            totalSteps = 0;
        }
    } else {
        totalSteps = 0;
    }
    http.end();
}

void updateRTDB(float lat, float lon) {
    FirebaseJson json;
    json.set("/latitude", String(lat, 6));
}
```



```

    json.set("/longitude", String(lon, 6));
    json.set("/Nav_Flag", Nav_Flag);
    if (Firebase.RTDB.updateNode(&fbdo, gpsDatabasePath.c_str(), &json)) {
        Serial.println("RTDB updated (GPS)");
    }
}

bool playInstruction(int stepNumber) {
    if (stepNumber > totalSteps || stepNumber < 1 || audioUrls[stepNumber - 1] == "") {
        return false;
    }
    audio.stopSong();
    int retries = 2;
    while (retries > 0 && !audio.connecttohost(audioUrls[stepNumber - 1].c_str())) {
        delay(500);
        audio.stopSong();
        retries--;
    }
    if (retries > 0) {
        while (audio.isRunning()) {
            audio.loop();
            delay(1);
        }
        Serial.println("Step " + String(stepNumber) + " audio played");
        return true;
    }
    return false;
}

void fetchNavFlag() {
    if (Firebase.ready()) {
        if (Firebase.RTDB.getBool(&fbdo, "/Sensors/gps/Nav_Flag")) {
            Nav_Flag = fbdo.boolData();
        }
    }
}

void updateFallStatus(String status) {
    Serial.print("Updating Fall Status in Firebase... ");
    if (Firebase.RTDB.setString(&fbdo, fallDatabasePath.c_str(), status)) {
        Serial.println("Success!");
    } else {
        Serial.println(fbdo.errorReason());
    }
}

int getDistance(int trigPin, int echoPin) {

```

```
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
long duration = pulseIn(echoPin, HIGH, 30000);
if (duration == 0) return -1;
int distance = duration * 0.034 / 2;
if (distance < DETECTION_MIN || distance > DETECTION_MAX) {
    return -1;
}
return distance;
}

void vibrateMotor(int pattern) {
    digitalWrite(VIB_PIN, HIGH);
    delay(VIB_DURATION);
    digitalWrite(VIB_PIN, LOW);
    delay(VIB_DURATION);
}
```

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